

Modeling and Forecasting Seasonality in International Air Travel

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Abstract

This project investigates the international airline passenger data from 1949 to 1960 to forecast future passenger trends using time series analysis. The main objective is to build an accurate forecasting model that can assist in aviation capacity planning and economic forecasting. After exploring the data and identifying non-constant variance and strong seasonality, transformations such as the logarithm and Box-Cox were applied to stabilize variance. Harmonic regression models were fitted to capture seasonal trends, with the log transformation and two harmonics ($k = 2$) showing the best fit (Adjusted $R^2 = 0.9763$). SARIMA models were used for better forecasting after diagnostic tests revealed non-stationarity in the residuals. AIC, BIC, and residual diagnostics were used to assess competing models. SARIMA(2,1,1)(0,1,1)[12], the finished model, produced precise forecasts and strong residual behaviour. Metrics like MSFE and MAFE were used to evaluate forecast performance, and rolling forecasts confirmed the model's dependability. Overall, a strong forecasting framework for the time series was created by combining the applications of transformation, harmonic regression, and SARIMA modelling.

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1 Introduction

This project uses time series analysis to forecast passenger demand for international airlines. The objective is to create a trustworthy model that aids in scheduling, resource allocation, and future travel trend preparation for airlines and policymakers. Predictions must be accurate: While underestimating demand strains infrastructure and irritates travellers, overestimating demand wastes money.

The data set used in this analysis is the well-known international airline passenger dataset, which contains monthly passenger counts (in thousands) from January 1949 to December 1960. By studying this dataset, we can evaluate how different time series models perform and how well they can predict future values.

A structured modeling method is used to address the problem. Section 2 provides an overview of the dataset and its characteristics. Section 3 details the model specification process, including exploratory data analysis, variance stabilization, harmonic regression, and preparation for time series modeling. Section 4 presents parameter estimation and diagnostic checks for candidate models. Forecasting procedures and model evaluation results are discussed in Section 5. Finally, Section 6 concludes the report with a discussion of the results and potential limitations.

2 Data Description

This project worked with the `AirPassengers` dataset, available by default in R. It tracks monthly international airline passenger numbers (in thousands) between January 1949 and December 1960. With 144 monthly records spanning 12 full years, the data shows noticeable upward trends and seasonal patterns.

The primary focus was on monthly passenger counts over this period. To understand the data's behavior, we first plotted the time series to visualize both the long-term growth

and recurring seasonal fluctuations.

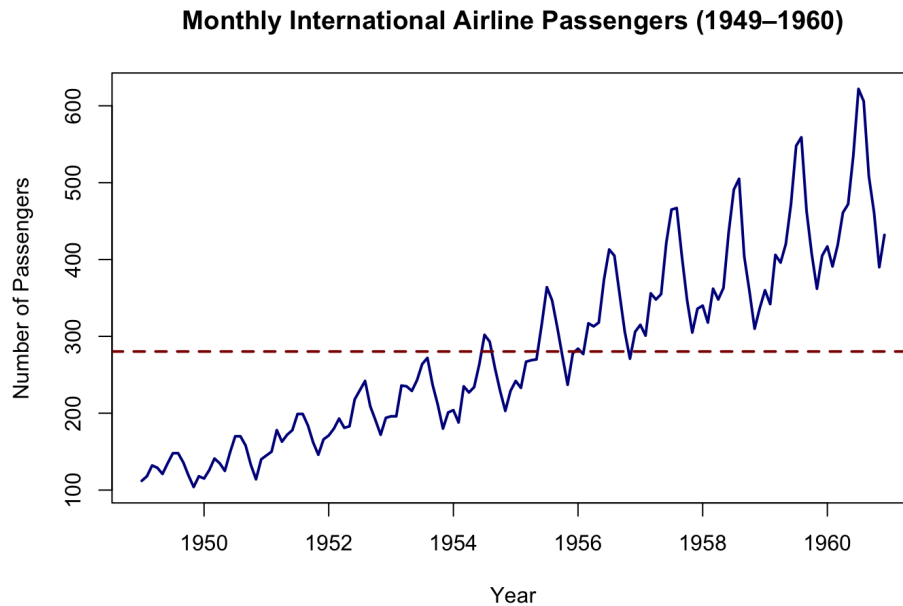


Figure 1: Monthly international airline passengers from 1949 to 1960.

3 Model Specification

The goal of model specification is to build an appropriate forecasting model by carefully analyzing the structure of the data. The process begins with an examination of the initial time plot to assess trend, seasonality, and variance behavior. Based on the observation of non-constant variance, transformations such as the Box-Cox and logarithmic transformations are applied.

Harmonic regression models are fitted to the converted series in order to address the data's strong seasonal pattern. After that, residual diagnostics are carried out to assess the model's suitability. ARMA modelling is used to capture the residual dependence structure because there is still a significant amount of autocorrelation in the residuals. The ensuing subsections provide a detailed description of each step.

3.1 Initial Exploration of Trend and Seasonality

An initial time plot (Figure 1) of the international airline passenger series reveals a clear upward trend over time, accompanied by noticeable seasonal fluctuations. The amplitude of these seasonal cycles appears to increase with the level of the series, suggesting the presence of non-constant variance (heteroscedasticity).

To further investigate the structure of the series, a classical decomposition was performed (Figure 2). This decomposition confirmed the presence of a strong trend and recurring seasonal patterns. Additionally, the residual component exhibited irregular fluctuations, reinforcing the need for transformation and model refinement.

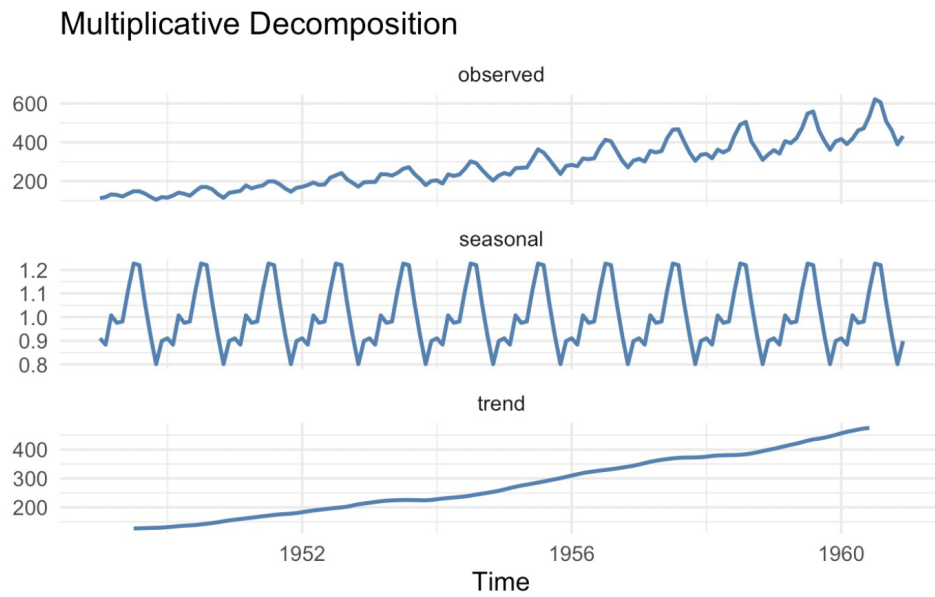


Figure 2: Classical decomposition of the AirPassengers time series.

3.2 Stabilizing the Variance

To stabilize the variance and better capture the trend and seasonal components, both log and Box-Cox transformations were applied. The Box-Cox analysis confirmed that a logarithmic transformation was suitable for the data.

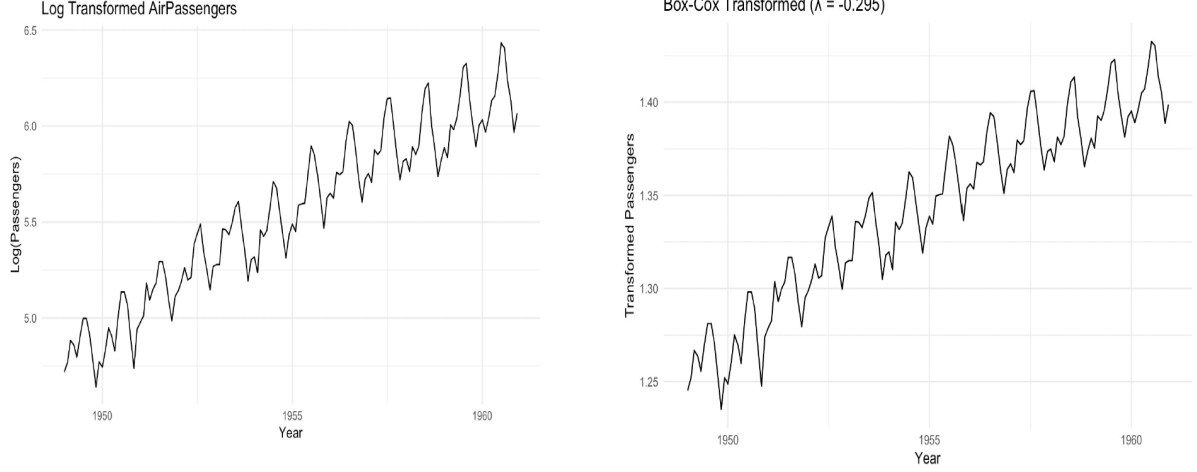


Figure 3: log and Box-Cox transformations

In order to address the strong seasonal patterns observed in the data, harmonic regression models were fit to both the log-transformed and Box-Cox-transformed series. This approach allowed for modeling the periodic structure using sine and cosine terms, while simultaneously accounting for the long-term trend. The evaluation and comparison of these models are discussed in the next section.

3.3 Harmonic Regression and Model Building

Both the log-transformed and Box-Cox-transformed series were subjected to harmonic regression in order to capture the data's strong seasonal behaviour. By employing periodic sine and cosine terms to model seasonality, harmonic regression enables flexible modelling of recurrent patterns without separating the data.

Models with one, two, and four harmonics were considered. In each case, the model was fit using a linear regression framework that included both trend terms and seasonal harmonics. The adjusted R^2 , residual standard error, and the statistical significance of the coefficient estimates were used to evaluate model performance.

A summary of the results is presented in Table 1. The log-transformed model with two harmonics achieved a high adjusted R^2 of 0.9763 and had all coefficients statistically

significant. The Box-Cox model with two harmonics also performed well, with a slightly lower residual standard error. While the models with four harmonics provided slightly better fit metrics, they introduced multicollinearity and included non-significant terms, reducing model interpretability.

Table 1: Comparison of Harmonic Regression Models

Transformation	Harmonics (k)	Adj R^2	Residual SE	Significant Terms?
Log	1	0.9589	0.0895	Yes
Log	2	0.9763	0.06803	Yes
Log	4	0.981	0.0609	No
Box-Cox	1	0.9500	0.01959	Yes
Box-Cox	2	0.9661	0.01612	Yes
Box-Cox	4	0.9707	0.015	No

The fitted values from the log-transformed model with two harmonics are shown in Figure 4. The model effectively captures the seasonal cycles and long-term trend in the data.

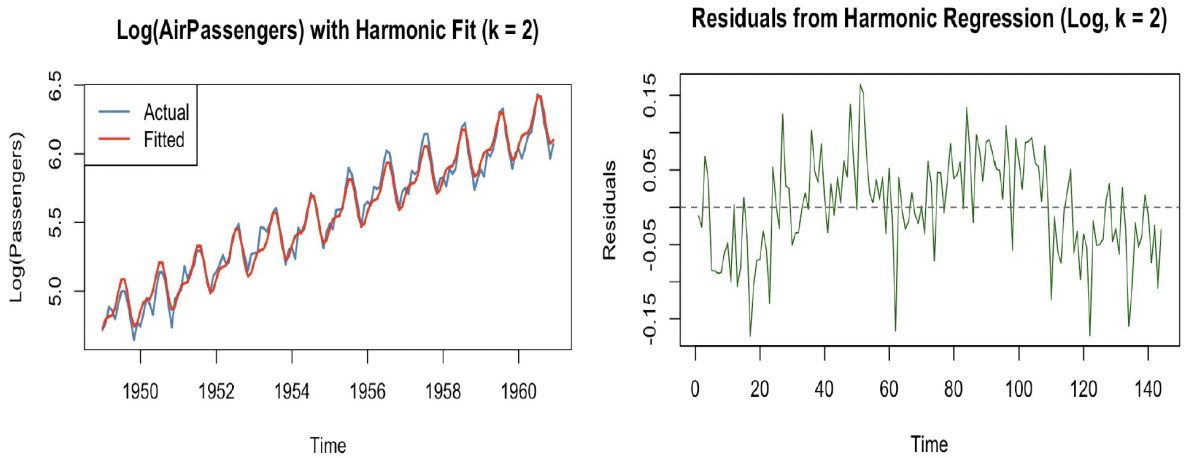


Figure 4: Fitted harmonic regression model with $k = 2$ harmonics on the log-transformed data.

Based on the comparison, the harmonic regression model with two seasonal components applied to the log-transformed data was selected for further analysis. In the following section, residuals from this model are examined in detail to assess its adequacy.

3.3.1 Diagnostic Tests on Harmonic Regression Residuals

To assess the adequacy of the harmonic regression model, diagnostic checks were performed on the residuals. These checks are essential for verifying that the remaining structure in the data is white noise and that the model has captured the systematic components effectively.

Stationarity: Augmented Dickey-Fuller Test

The Augmented Dickey-Fuller (ADF) test was used to assess the stationarity of the residuals. For both the log-transformed and Box-Cox-transformed models, the test failed to reject the null hypothesis of a unit root, indicating that the residuals were not stationary.

- Log-transformed model: ADF p-value = 0.305
- Box-Cox model: ADF p-value = 0.764

Whiteness: Box-Ljung Test

The Box-Ljung test was applied to examine whether the residuals resemble white noise. For both models, the null hypothesis was strongly rejected, suggesting that autocorrelation remains in the residuals.

- Log-transformed model: p-value $< 2.2 \times 10^{-16}$
- Box-Cox model: p-value $< 2.2 \times 10^{-16}$

Normality: Shapiro-Wilk Test

To assess whether the residuals follow a normal distribution, the Shapiro-Wilk test was conducted. Both models produced p-values above the 0.05 threshold, indicating that

normality could not be rejected.

- Log-transformed model: p-value = 0.733
- Box-Cox model: p-value = 0.127

Normality: Q-Q Plot Inspection

In addition to the Shapiro-Wilk test, Q-Q plots were used to visually assess the normality of residuals from the harmonic regression models with $k = 2$. As shown in Figure 5, both the log-transformed and Box-Cox-transformed residuals follow the theoretical quantile line reasonably well, indicating approximate normality.

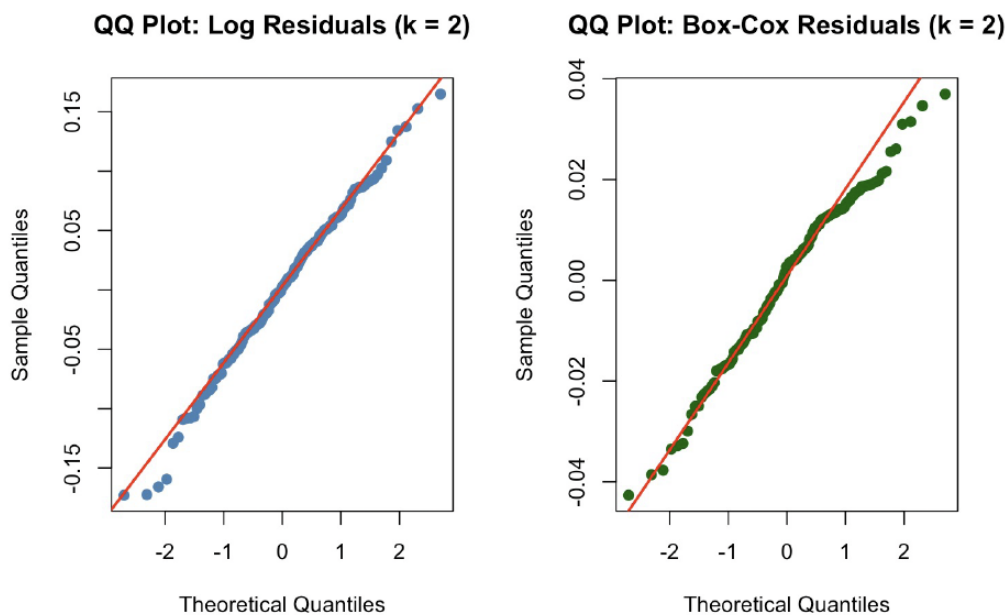


Figure 5: Q-Q plots of residuals from harmonic regression models with $k = 2$: log-transformed (left) and Box-Cox-transformed (right).

Although both models produced reasonably normal residuals, the Q-Q plot for the log-transformed residuals shows closer alignment with the theoretical line. Based on this, the log-transformed model was selected for further modeling.

The results suggest that while the residuals are approximately normal, they are neither

stationary nor white noise. These findings motivate the use of time series modeling techniques, such as ARMA or SARIMA, to capture the remaining autocorrelation structure.

3.4 Achieving Stationarity

Before fitting ARMA models, it is necessary to ensure that the residual series is stationary, as stationarity is a fundamental assumption for ARMA modeling. The residuals from the harmonic regression were initially non-stationary, as indicated by diagnostic tests and visual inspection. To address this, the residuals were differenced once.

After differencing, several diagnostic tests were conducted to confirm stationarity and randomness:

- **Augmented Dickey-Fuller (ADF) Test:** The ADF test statistic was -6.6956 with a p-value of 0.01, allowing us to reject the null hypothesis of a unit root. This provides strong evidence that the differenced residuals are stationary.
- **KPSS Test:** The KPSS test for level stationarity produced a test statistic of 0.02765 with a p-value of 0.1, further supporting stationarity.
- **Box-Ljung Test:** The Box-Ljung test yielded a test statistic of 235.68 with a p-value less than 2.2×10^{-16} , indicating the presence of significant autocorrelation remaining in the series.
- **Runs Test:** The observed number of runs was 88 compared to an expected 72.08, with a p-value of 0.00891, suggesting some deviation from perfect randomness.

A summary of these results is provided in Table 2.

Table 2: Diagnostic Test Results for Differenced Residuals

Test	Test Statistic	p-value
Augmented Dickey-Fuller Test	-6.6956	0.01
KPSS Test	0.02765	0.1
Box-Ljung Test	235.68	$< 2.2 \times 10^{-16}$
Runs Test	Observed: 88, Expected: 72.08	0.00891

Overall, the differenced residuals exhibited sufficient stationarity for ARMA modeling. However, due to the significant autocorrelation identified by the Box-Ljung test, further modeling of the series using ARMA models was necessary. The model specification and selection are discussed in the next subsection.

3.4.1 Model Order Identification: ARMA Subset Selection

The Extended Autocorrelation Function (EACF) table (Table 3) was used to assist in selecting candidate ARMA models. Based on the EACF results, several models, including ARMA(0,3), ARMA(1,3), ARMA(2,3), and ARMA(2,2), were identified for further consideration.

Table 3: Extended Autocorrelation Function (EACF) Table for Differenced Residuals

AR/MA	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	x	x	x	o	o	o	o	o	x	x	x	x	x	x
1	x	x	x	o	o	o	o	o	x	o	x	x	x	x
2	o	x	o	o	o	o	o	o	o	o	o	x	o	x
3	x	x	o	o	o	o	o	o	o	o	o	x	o	o
4	o	x	o	o	o	o	o	o	o	o	o	x	x	o
5	o	x	o	o	o	x	o	o	o	o	o	x	x	o
6	x	x	o	x	o	x	o	o	o	o	o	x	o	o
7	x	x	o	x	o	x	o	o	o	o	o	x	o	o

3.5 Diagnostic Checks for ARMA Models

After fitting a set of ARMA models to the differenced residuals, a series of diagnostic tests were performed to evaluate model adequacy. These included the analysis of standardized

residuals, autocorrelation through ACF plots, and formal statistical tests such as the Ljung-Box test for residual independence, the runs test for randomness, and the Shapiro-Wilk test for normality.

While the diagnostic plots generated by `tsdiag()` (Figure 6) appeared acceptable—showing residuals mostly within bounds and Ljung-Box p-values above the 5% significance level—the formal test results suggested otherwise. Specifically, several models exhibited statistically significant residual autocorrelation when the Ljung-Box test was applied up to lag 20 using `Box.test()`. This discrepancy highlighted the limitations of relying solely on visual diagnostics and underscored the need for a more robust assessment using formal tests.

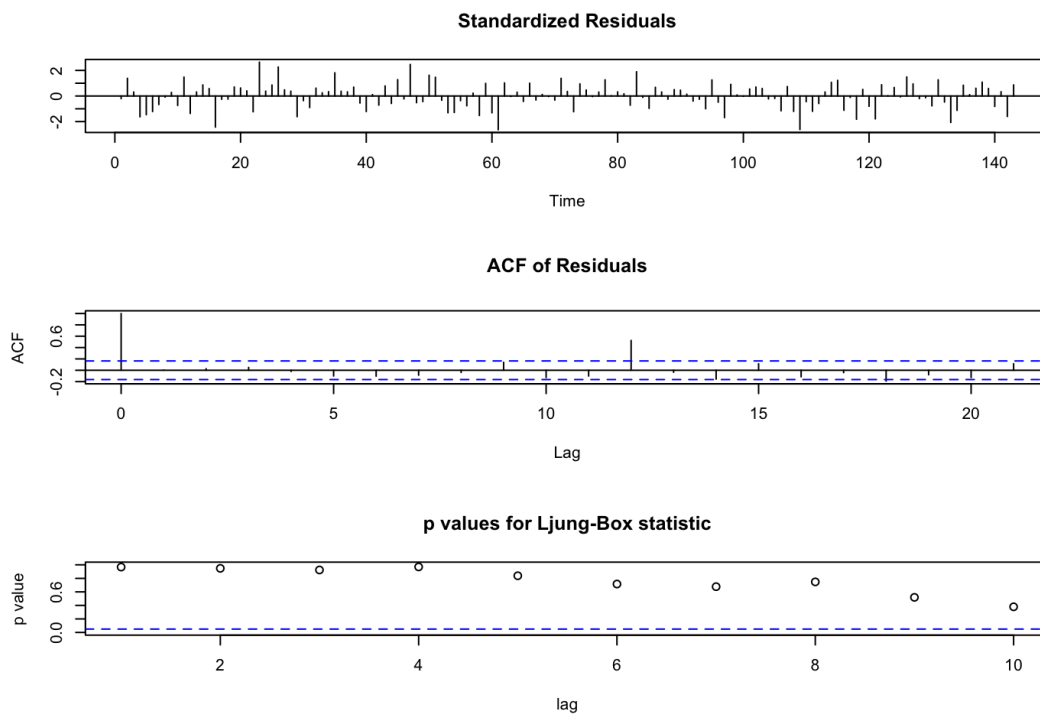


Figure 6: Residual diagnostics from `tsdiag()` for the ARMA(2,2) model

The table below summarizes the diagnostic test results for the four ARMA models under consideration:

Table 4: Diagnostic Test Results for Fitted ARMA Models

Model	AIC	BIC	Ljung-Box p	Runs Test p	Shapiro-Wilk p
ARMA(0,3)	-409.19	-394.38	6.51×10^{-13}	0.1778	0.8579
ARMA(1,3)	-414.35	-396.57	3.89×10^{-9}	0.5005	0.7369
ARMA(2,3)	-413.39	-392.65	3.64×10^{-8}	0.5005	0.6727
ARMA(2,2)	-415.23	-397.45	3.28×10^{-8}	0.5005	0.6503

Although ARMA(2,2) produced the lowest AIC and BIC values, none of the models passed the Ljung-Box test at higher lags. This indicated remaining autocorrelation in the residuals, motivating a shift toward SARIMA modeling to explicitly account for seasonality.

3.6 Preparing for SARIMA Modeling

Given the presence of residual autocorrelation and suspected seasonal effects in the data, SARIMA modeling was pursued to better capture both non-seasonal and seasonal dependencies. To prepare the residual series for SARIMA analysis, two differencing operations were applied. First, a seasonal difference at lag 12 was used to remove recurring yearly patterns. Then, a first-order non-seasonal difference was applied to eliminate any remaining trend in the data. This resulted in a twice-differenced series, denoted as `final_diff`, which was then subjected to diagnostic testing.

The Augmented Dickey-Fuller (ADF) test on `final_diff` yielded a Dickey-Fuller statistic of -5.1993 with a p-value of 0.01, allowing us to reject the null hypothesis of a unit root. This confirmed that the doubly differenced series is stationary.

In addition, the Box-Ljung test was applied to evaluate residual autocorrelation. The test yielded a statistic of 64.598 with a p-value of 1.352×10^{-6} , indicating that significant autocorrelation remains in the series and justifying the need for further modeling.

- **ADF Test (stationarity):** statistic = -5.1993, p-value = 0.01

- **Box-Ljung Test (lag 20):** statistic = 64.598, p-value = 1.352×10^{-6}

Based on these results, SARIMA modeling was pursued to account for both the seasonal and non-seasonal dynamics in the residuals.

3.6.1 Model Selection for SARIMA

To identify appropriate SARIMA models for the differenced residual series, the extended autocorrelation function (EACF) was used. The EACF table below shows potential combinations of AR and MA orders based on the placement of "o" values.

Table 5: EACF Table for Differenced Residuals (lag = 12)

AR/MA	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	x	o	x	o	o	o	o	o	o	o	x	o	o	o
1	o	x	x	o	o	o	o	o	o	o	x	o	o	o
2	o	x	x	o	o	o	o	o	o	o	x	o	o	o
3	x	x	x	o	o	o	o	o	o	o	x	o	o	o
4	x	x	x	o	o	o	o	o	o	o	x	o	o	o
5	x	x	x	o	o	o	o	o	o	o	x	o	o	o
6	x	x	x	o	o	o	o	o	o	o	x	o	o	o

Based on the EACF table, the following SARIMA models were selected for further evaluation:

- SARIMA(0,1,1)(0,1,1)[12]
- SARIMA(1,1,1)(0,1,1)[12]
- SARIMA(2,1,1)(0,1,1)[12]
- SARIMA(1,1,2)(0,1,1)[12]
- SARIMA(0,1,3)(0,1,1)[12]
- SARIMA(1,1,3)(0,1,1)[12]

These candidate models will be compared using information criteria and residual diagnostics to determine the best-performing SARIMA model.

4 Parameter Estimation and Diagnostic Check

Each of the candidate SARIMA models identified through EACF analysis was fitted to the transformed series and evaluated using multiple diagnostic criteria. These included:

- **Model selection metrics:** Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC)
- **Coefficient significance:** Based on the t-values and p-values of estimated parameters
- **Residual diagnostics:** Box-Ljung test for autocorrelation, Shapiro-Wilk test for normality, and the runs test for randomness

The results are summarized in Table 6. Coefficient significance was assessed through the model summaries, while diagnostic test results were interpreted using a standard significance threshold of 0.05.

Table 6: Diagnostic Results for Candidate SARIMA Models

Model	AIC	BIC	Significant Terms	Box-Ljung	Shapiro-Wilk	Runs Test
SARIMA(0,1,1)(0,1,1)[12]	-481.89	-470.39	Yes	0.623	0.140	0.145
SARIMA(1,1,1)(0,1,1)[12]	-485.89	-470.39	No	0.687	0.067	0.416
SARIMA(2,1,1)(0,1,1)[12]	-482.26	-467.88	Yes	0.577	0.019	0.400
SARIMA(1,1,2)(0,1,1)[12]	-484.03	-467.65	Yes	0.676	0.025	0.435
SARIMA(0,1,3)(0,1,1)[12]	-482.63	-468.26	No	0.600	0.133	0.386
SARIMA(1,1,3)(0,1,1)[12]	-480.81	-463.56	No	0.595	0.101	0.386

Although SARIMA(1,1,1)(0,1,1)[12] achieved the lowest AIC value, its coefficients were not statistically significant, making it unsuitable for further analysis.

SARIMA(2,1,1)(0,1,1)[12] and SARIMA(1,1,2)(0,1,1)[12] had significant terms and generally passed most residual diagnostics, although their Shapiro-Wilk p-values were slightly below the 0.05 threshold.

In addition to standard residual diagnostics, ARCH tests were conducted on the squared residuals of each candidate SARIMA model to assess the presence of conditional heteroscedasticity. The ARCH LM-tests were performed using 12 lags. The resulting p-values were as follows:

- SARIMA(2,1,1)(0,1,1)[12]: p-value = 0.2502
- SARIMA(1,1,2)(0,1,1)[12]: p-value = 0.5523
- SARIMA(0,1,1)(0,1,1)[12]: p-value = 0.5549

Since all p-values were greater than 0.05, we failed to reject the null hypothesis of no ARCH effects for all three models. This indicates that there was no significant evidence of conditional heteroscedasticity in the residuals, and thus no need for GARCH-type modeling.

It is important to note that the Shapiro-Wilk test is known to be sensitive to outliers. Therefore, in addition to relying on the formal test results, Q-Q plots were examined to assess residual normality visually.

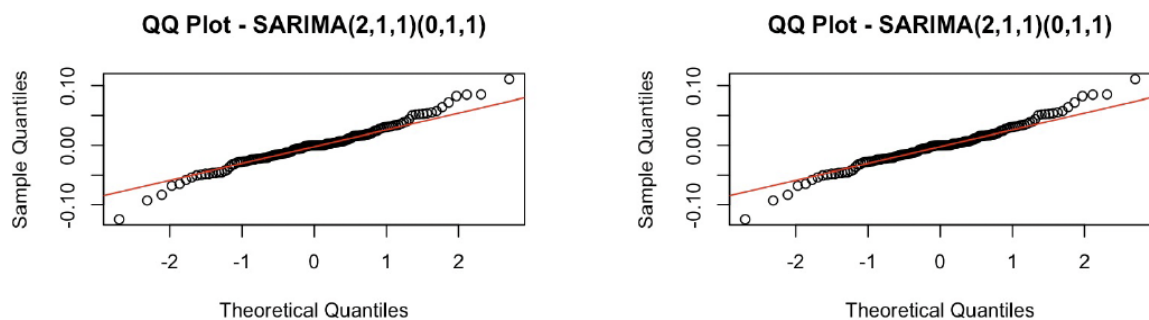


Figure 7: Q-Q plots of residuals for SARIMA(1,1,2) and SARIMA(2,1,1) models.

As shown in Figure 7, the residuals from both SARIMA(2,1,1)(0,1,1)[12] and SARIMA(1,1,2)(0,1,1)[12] closely follow a normal distribution, with only minor deviations in the tails. This confirms that the assumption of normality is reasonably satisfied for both models, despite the Shapiro-Wilk test returning p-values slightly below 0.05.

Additionally, SARIMA(0,1,1)(0,1,1)[12] demonstrated acceptable residual behavior, passing all diagnostic tests and yielding significant coefficients. However, its AIC and BIC values were slightly higher compared to the other two models.

Models Selected for Final Evaluation and Forecasting

Based on residual diagnostics, coefficient significance, and model fit criteria, the following SARIMA models were selected for final evaluation and forecasting:

- SARIMA(0,1,1)(0,1,1)[12]
- SARIMA(2,1,1)(0,1,1)[12]
- SARIMA(1,1,2)(0,1,1)[12]

5 Forecasting

Forecasting plays a critical role in time series analysis, particularly in applications involving planning, capacity management, and policy decisions. In this study, the final models selected in the previous section were used to generate forecasts for the next 24 months of international airline passenger volumes.

5.1 Forecasting Setup

To evaluate predictive performance, the final 24 months of the dataset were held out as a test set, with the remaining data used for model fitting. Each model was used to generate forecasts on this test set using a rolling 1-step-ahead forecasting strategy. In this approach, the model is refit as each new observation becomes available, and a one-step forecast is made at each point.

5.2 Model Evaluation Strategy

Forecast accuracy was assessed using the following evaluation metrics:

- Mean Squared Forecast Error (MSFE)
- Mean Absolute Forecast Error (MAFE)
- Diebold–Mariano Test

MSFE penalizes large forecast errors by squaring them, while MAFE provides a more interpretable average absolute error. The Diebold–Mariano test is a statistical procedure used to compare the predictive accuracy of two competing models.

Table 7: Forecast Accuracy Metrics for Final SARIMA Models

Model	Test MSFE	Test MAFE	Rolling MSFE	Rolling MAFE
SARIMA(0,1,1)(0,1,1)[12]	0.00920	0.08960	0.00122	0.02647
SARIMA(2,1,1)(0,1,1)[12]	0.00135	0.03182	0.00125	0.02686
SARIMA(1,1,2)(0,1,1)[12]	0.00645	0.07411	0.00127	0.02696

As shown in Table 7, SARIMA(2,1,1)(0,1,1)[12] outperformed the other models in terms of both test set and rolling forecast accuracy. Figure 8 shows the 24-month forecast produced by this model, along with prediction intervals that capture the uncertainty in future values.

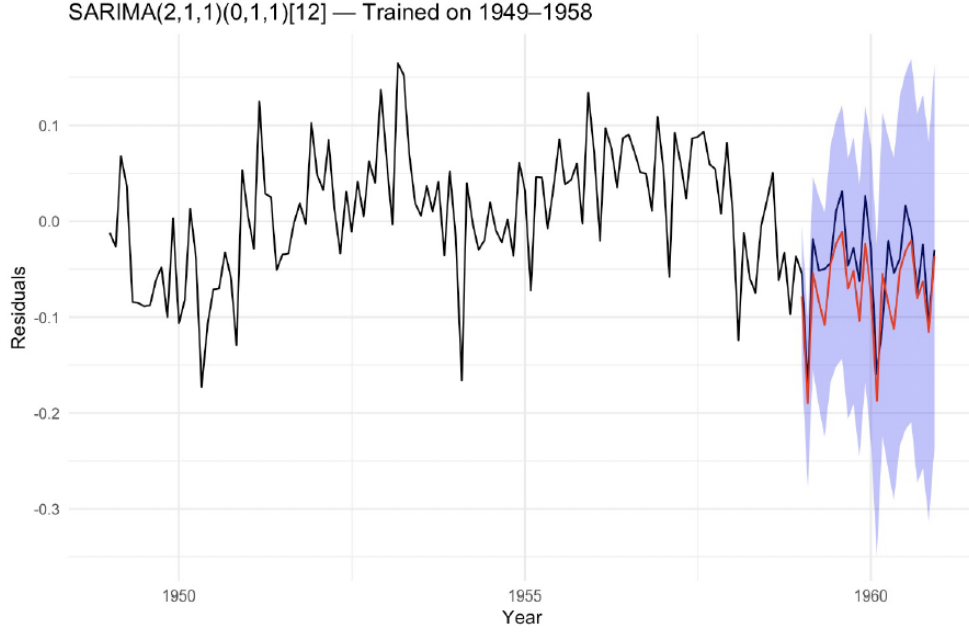


Figure 8: 24-month forecast from SARIMA(2,1,1)(0,1,1)[12]

The Diebold–Mariano (DM) test was used to statistically assess whether the forecast errors from SARIMA(2,1,1)(0,1,1)[12] were significantly different from those of SARIMA(0,1,1)(0,1,1)[12]. The test compared the squared forecast errors from each model over the 24-month test set.

The resulting DM statistic was -6.94 with a p-value less than 0.001, indicating strong evidence against the null hypothesis of equal predictive accuracy. This confirms that SARIMA(2,1,1)(0,1,1)[12] provides significantly better forecasts than the other model.

Given its strong performance across all evaluation metrics and its stable forecast behavior, SARIMA(2,1,1)(0,1,1)[12] was selected as the final model for forecasting international airline passenger volumes.

5.3 Generating the Final Forecast for 1961

After selecting SARIMA(2,1,1)(0,1,1)[12] as the final model, the following steps were taken to generate the final forecast:

1. The harmonic regression model was refit to the full log-transformed dataset (1949–1960).
2. Residuals were extracted from this full harmonic model.
3. SARIMA(2,1,1)(0,1,1)[12] was then fit to these residuals.
4. Forecasted residuals were added to the fitted harmonic trend to produce the final log-scale forecast for 1961.

Figure 9 illustrates the final forecasting process. The left panel displays the forecasted residuals obtained from the SARIMA(2,1,1)(0,1,1)[12] model. The right panel shows the resulting forecast for 1961, produced by adding these forecasted residuals to the fitted harmonic trend model. The forecast is shown on the log scale and includes both trend and seasonal components, resulting in a smooth, interpretable projection.

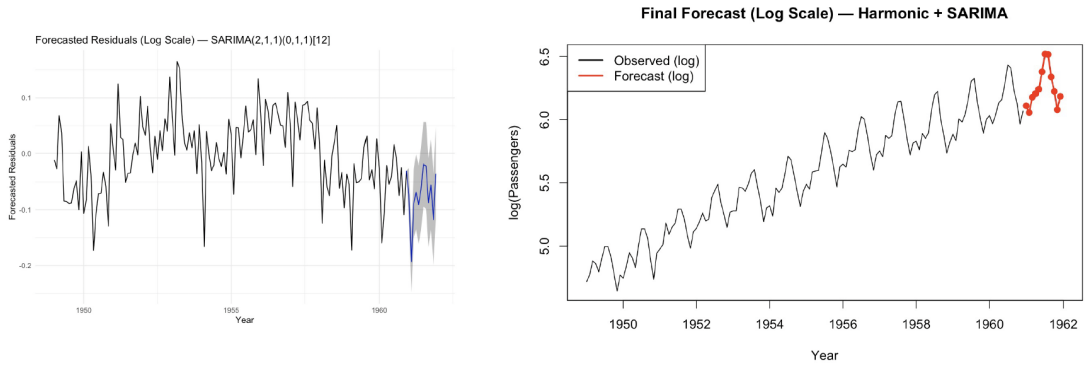


Figure 9: Final forecasting

6 Conclusion

This analysis explored forecasting monthly international airline passengers using a two-step approach: harmonic regression to capture trend and seasonality, and SARIMA modeling to handle residual autocorrelation. The data showed clear trend, strong seasonality, and increasing variance, which were addressed using a log transformation and harmonic terms.

Residuals from the harmonic model were modeled using SARIMA(2,1,1)(0,1,1)[12], selected based on information criteria (AIC), forecast error metrics (MSFE, MAFE), and

the Diebold–Mariano test. Residual checks indicated no major autocorrelation or evidence of GARCH effects, supporting the use of a SARIMA model without conditional heteroscedasticity.

The final forecast for 1961 was produced by combining the fitted harmonic trend with the forecasted residuals, yielding a log-scale forecast that captures both structural and dynamic components of the time series.

Limitations and Future Work

While the approach was effective, several limitations should be noted. First, the forecast remains on the log scale, and no back-transformation was applied, which may limit direct interpretability. Additionally, because no actual passenger data from 1961 was available, the forecasts could not be validated on truly unseen future values.

Future work could focus on forecasting on the original scale, incorporating confidence intervals after back-transformation, or exploring alternative modeling frameworks.

7 References

- Cryer, J. D., and Chan, K.-S. (2008). *Time Series Analysis with Applications in R* (2nd ed.). Springer. Available electronically through the Wilfrid Laurier University Library.
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Air Passenger

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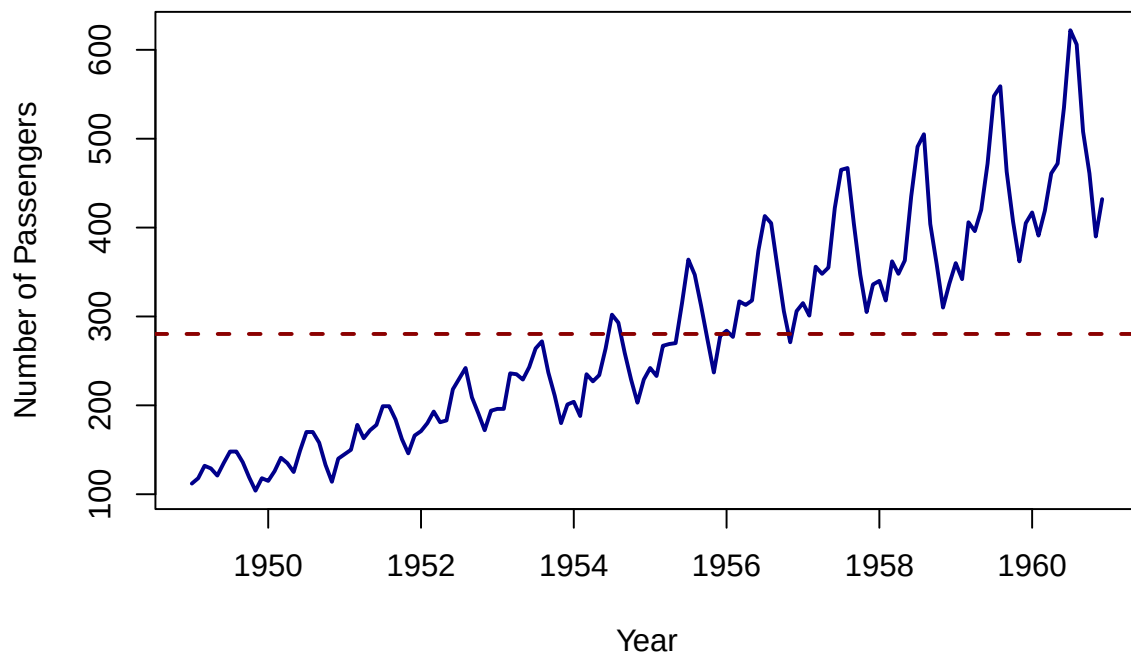
2025-03-27

```
data("AirPassengers")

plot(AirPassengers,
     main = "Monthly International Airline Passengers (1949-1960)",
     ylab = "Number of Passengers",
     xlab = "Year",
     col = "darkblue",
     lwd = 2)

abline(h = mean(AirPassengers), col = "darkred", lty = "dashed", lwd = 2)
```

Monthly International Airline Passengers (1949–1960)



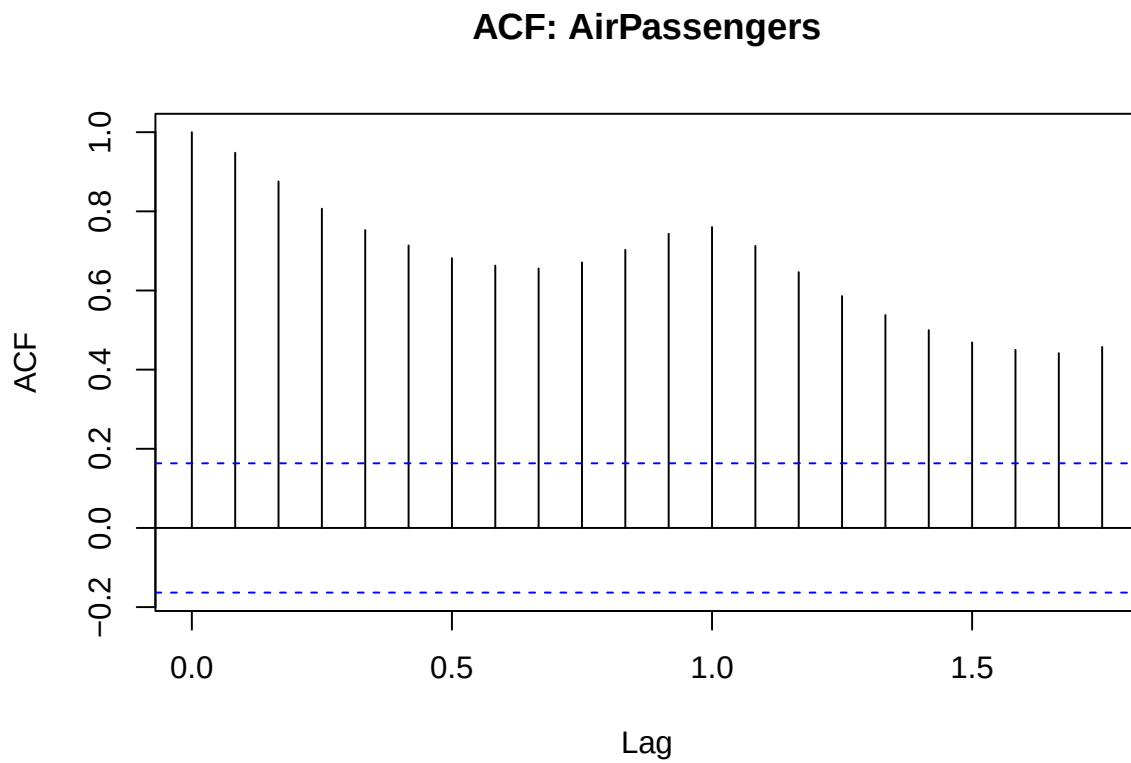
```
library(forecast)

## Registered S3 method overwritten by 'quantmod':
##   method      from
## as.zoo.data.frame zoo
```

```
library(tseries)

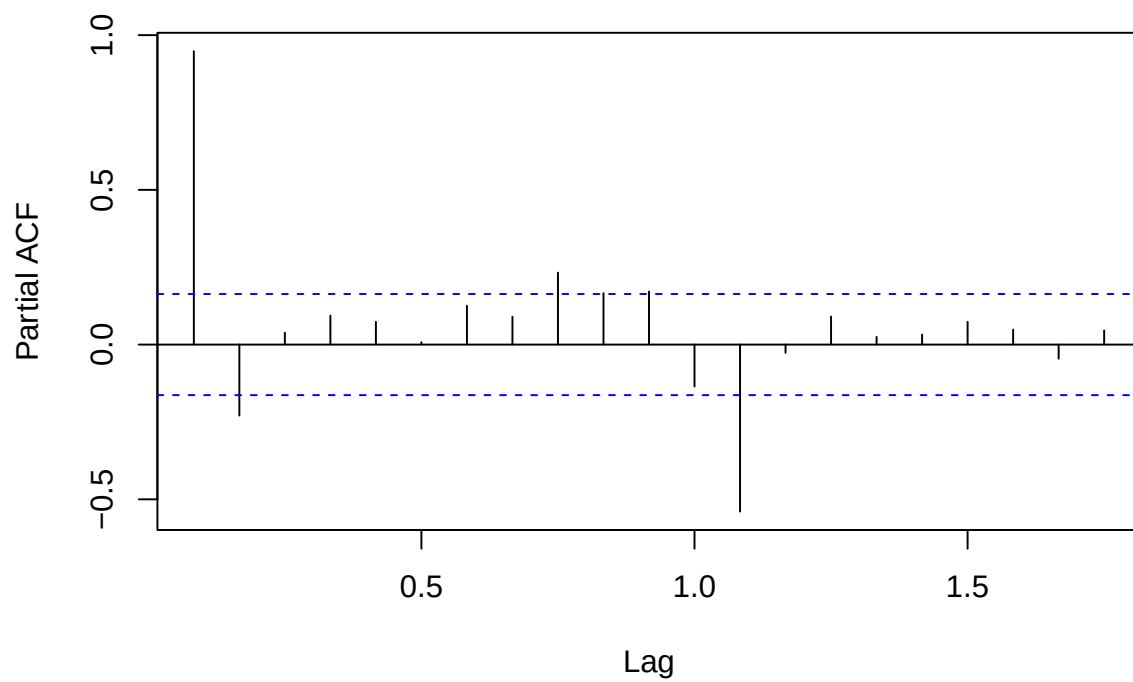
# ACF and PACF of the original AirPassengers data

acf(AirPassengers, main = "ACF: AirPassengers")
```



```
pacf(AirPassengers, main = "PACF: AirPassengers")
```

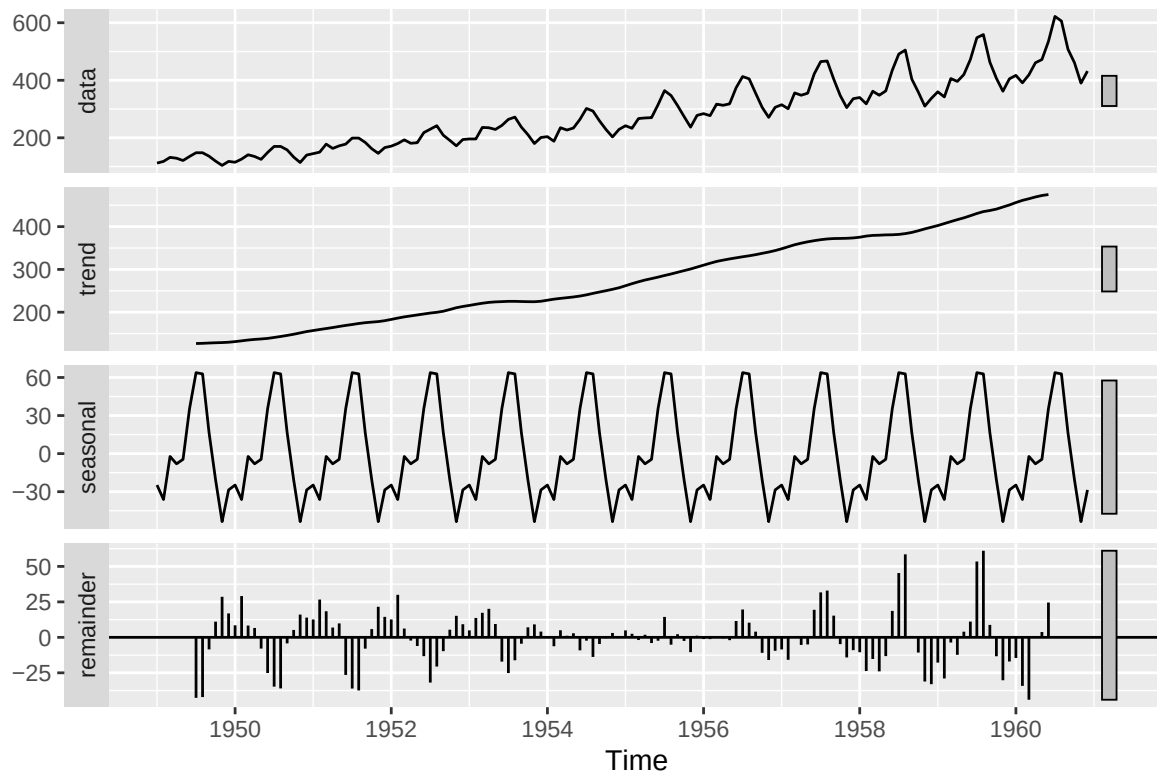
PACF: AirPassengers



```
decomp <- decompose(AirPassengers)

# Plot the decomposed components
autoplot(decomp)
```

Decomposition of additive time series



```
library(ggplot2)
library(ggfortify)
```

```
## Registered S3 methods overwritten by 'ggfortify':
```

## method	from
## autoplot.Arima	forecast
## autoplot.acf	forecast
## autoplot.ar	forecast
## autoplot.bats	forecast
## autoplot.decomposed.ts	forecast
## autoplot.ets	forecast
## autoplot.forecast	forecast
## autoplot.stl	forecast
## autoplot.ts	forecast
## fitted.ar	forecast
## fortify.ts	forecast
## residuals.ar	forecast

```
library(forecast)
library(TSA)
```

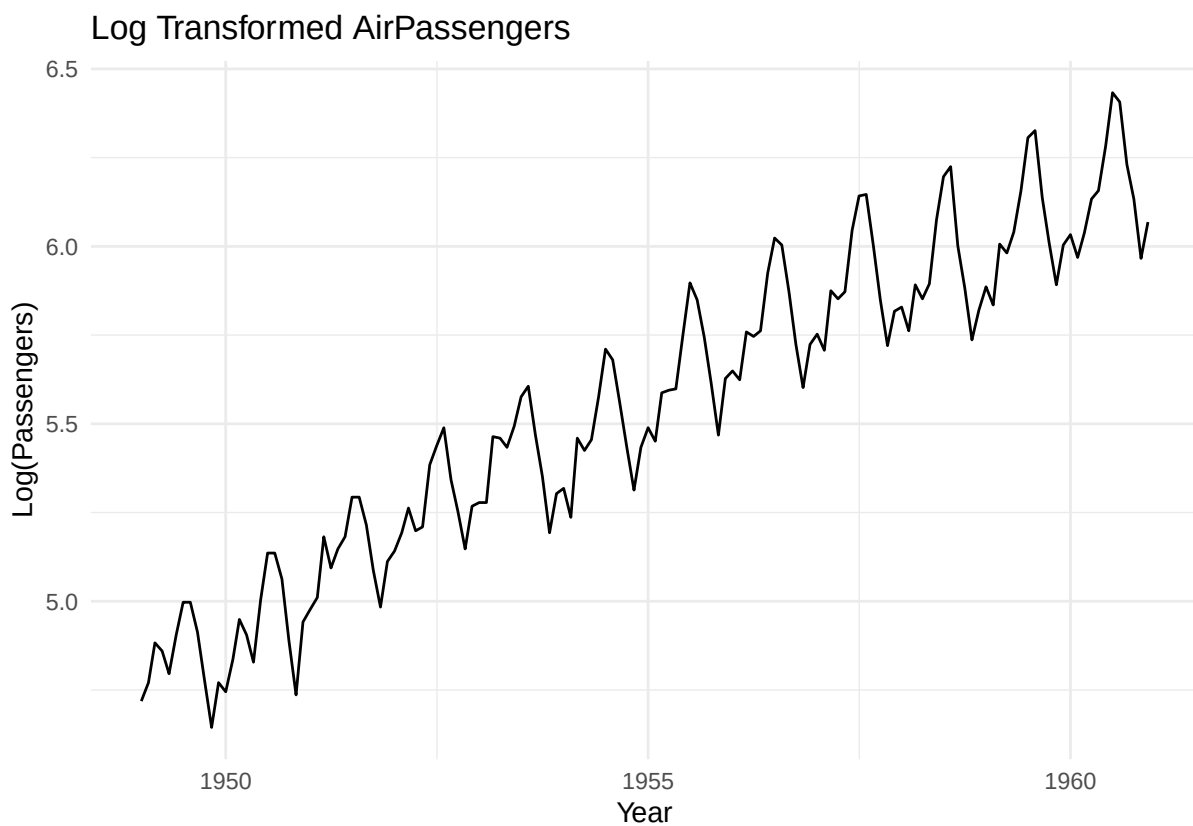
```
## Registered S3 methods overwritten by 'TSA':
```

## method	from
## fitted.Arima	forecast
## plot.Arima	forecast

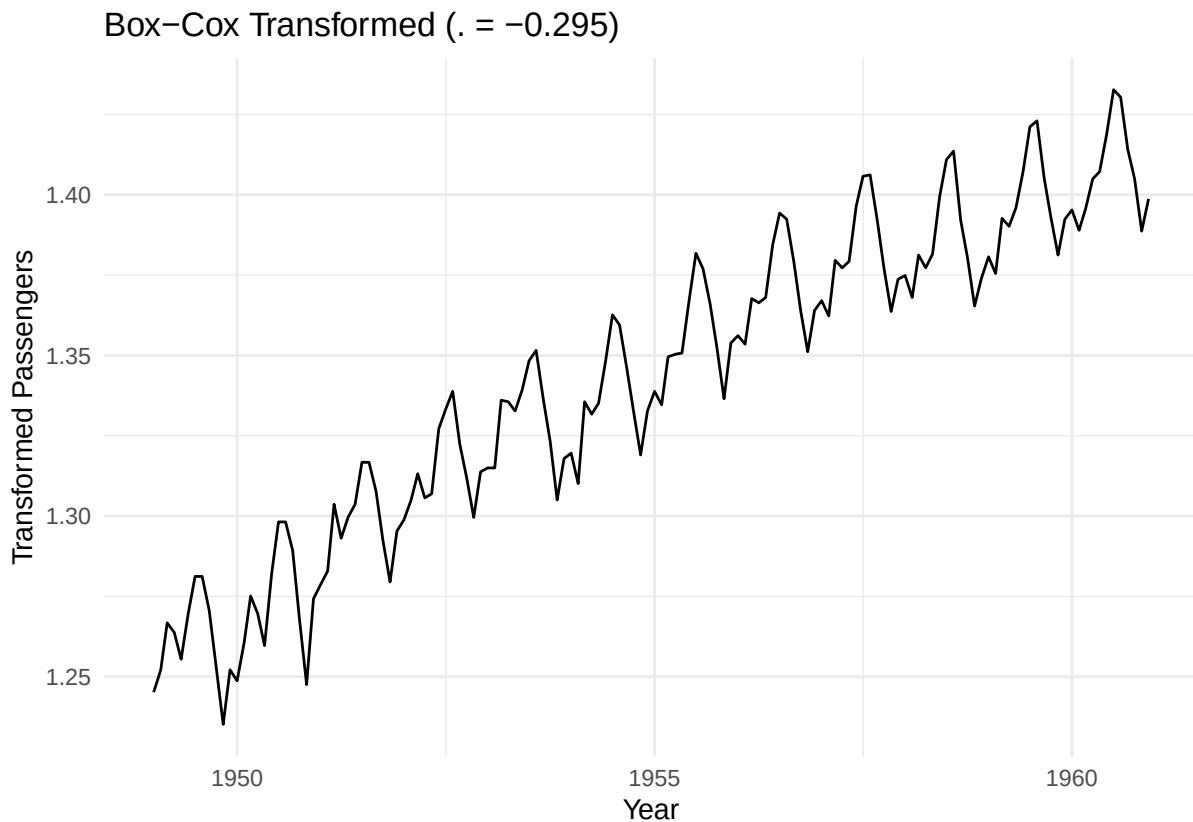
```
##
```

```
## Attaching package: 'TSA'
```

```
## The following objects are masked from 'package:stats':
##
##   acf, arima
##
## The following object is masked from 'package:utils':
##
##   tar
# Log transform
log_ap_ts <- ts(log(AirPassengers), start = c(1949, 1), frequency = 12)
autoplot(log_ap_ts) +
  ggtitle("Log Transformed AirPassengers") +
  ylab("Log(Passengers)") +
  xlab("Year") +
  theme_minimal()
```



```
# Box-Cox transform
lambda <- -0.2947156
ts_boxcox <- BoxCox(log_ap_ts, lambda)
ts_boxcox <- ts(ts_boxcox, start = c(1949, 1), frequency = 12)
autoplot(ts_boxcox) +
  ggtitle(paste0("Box-Cox Transformed ( = ", round(lambda, 3), ")")) +
  ylab("Transformed Passengers") +
  xlab("Year") +
  theme_minimal()
```



```
library(ggplot2)
library(ggfortify)
library(forecast)
library(TSA)

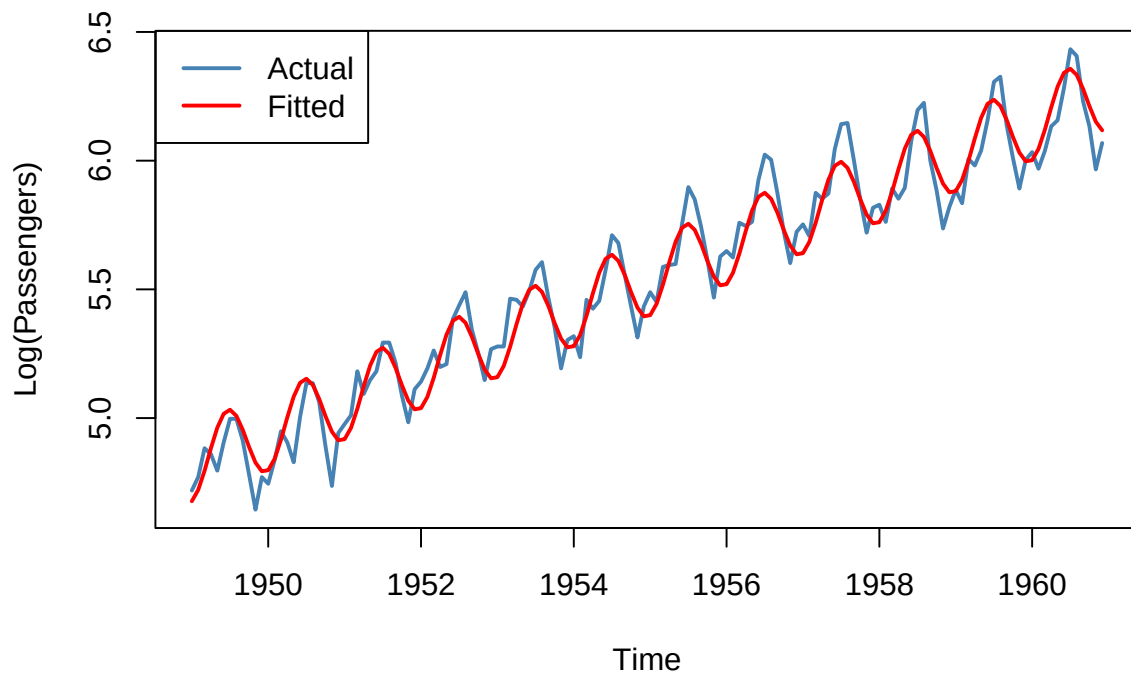
log_ap_ts <- ts(log(AirPassengers), start = c(1949, 1), frequency = 12)

lambda <- BoxCox.lambda(log_ap_ts)
ts_boxcox <- BoxCox(log_ap_ts, lambda)
ts_boxcox <- ts(ts_boxcox, start = c(1949, 1), frequency = 12)

time_vals <- time(log_ap_ts)
harm_model_log_k1 <- lm(log_ap_ts ~ time_vals + harmonic(log_ap_ts, 1))
fitted_log_k1 <- ts(fitted(harm_model_log_k1), start = start(log_ap_ts), frequency = frequency(log_ap_ts))
resid_log_k1 <- residuals(harm_model_log_k1)

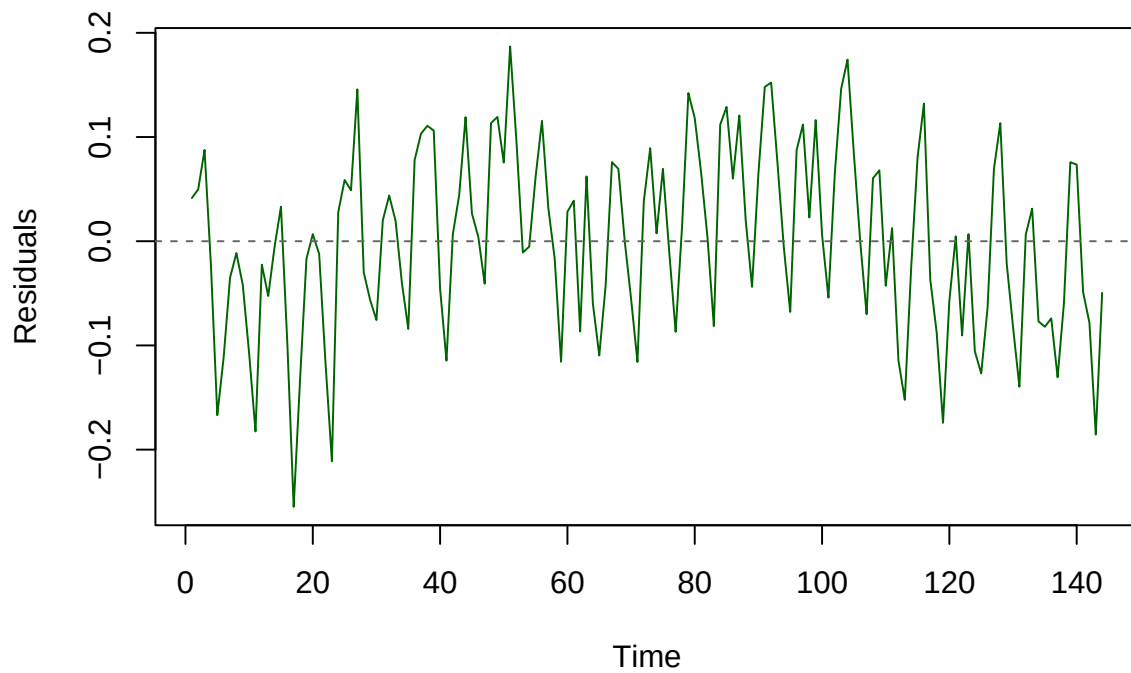
plot(log_ap_ts, type = "l", col = "steelblue", lwd = 2,
     ylab = "Log(Passengers)", xlab = "Time",
     main = "Log(AirPassengers) with Harmonic Fit (k = 1)")
lines(fitted_log_k1, col = "red", lwd = 2)
legend("topleft", legend = c("Actual", "Fitted"), col = c("steelblue", "red"), lwd = 2)
```

Log(AirPassengers) with Harmonic Fit (k = 1)



```
plot(resid_log_k1, type = "l", col = "darkgreen",  
     main = "Residuals from Harmonic Regression (Log, k = 1)",  
     ylab = "Residuals", xlab = "Time")  
abline(h = 0, lty = "dashed", col = "gray40")
```

Residuals from Harmonic Regression (Log, k = 1)



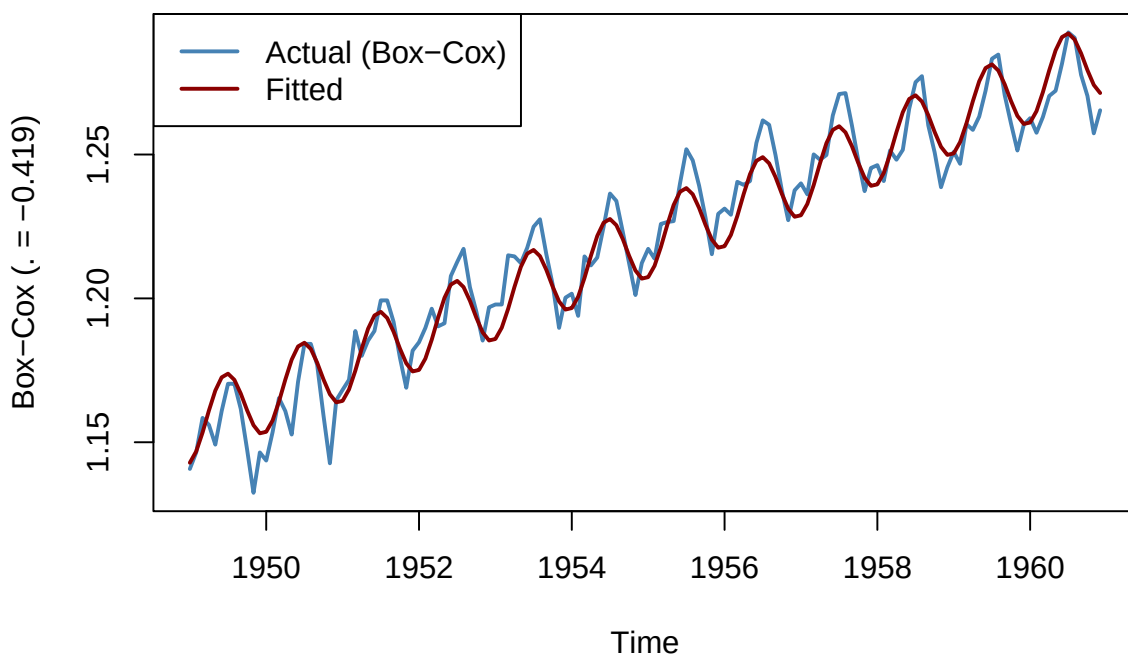
```

time_vals_bc <- time(ts_boxcox)
harm_model_bc_k1 <- lm(ts_boxcox ~ time_vals_bc + harmonic(ts_boxcox, 1))
fitted_bc_k1 <- ts(fitted(harm_model_bc_k1), start = start(ts_boxcox), frequency = frequency(ts_boxcox))
resid_bc_k1 <- residuals(harm_model_bc_k1)

plot(ts_boxcox, type = "l", col = "steelblue", lwd = 2,
     ylab = paste0("Box-Cox ( = ", round(lambda, 3), " )"), xlab = "Time",
     main = paste0("Harmonic Regression with Box-Cox (k = 1)"))
lines(fitted_bc_k1, col = "darkred", lwd = 2)
legend("topleft", legend = c("Actual (Box-Cox)", "Fitted"), col = c("steelblue", "darkred"), lwd = 2)

```

Harmonic Regression with Box-Cox (k = 1)

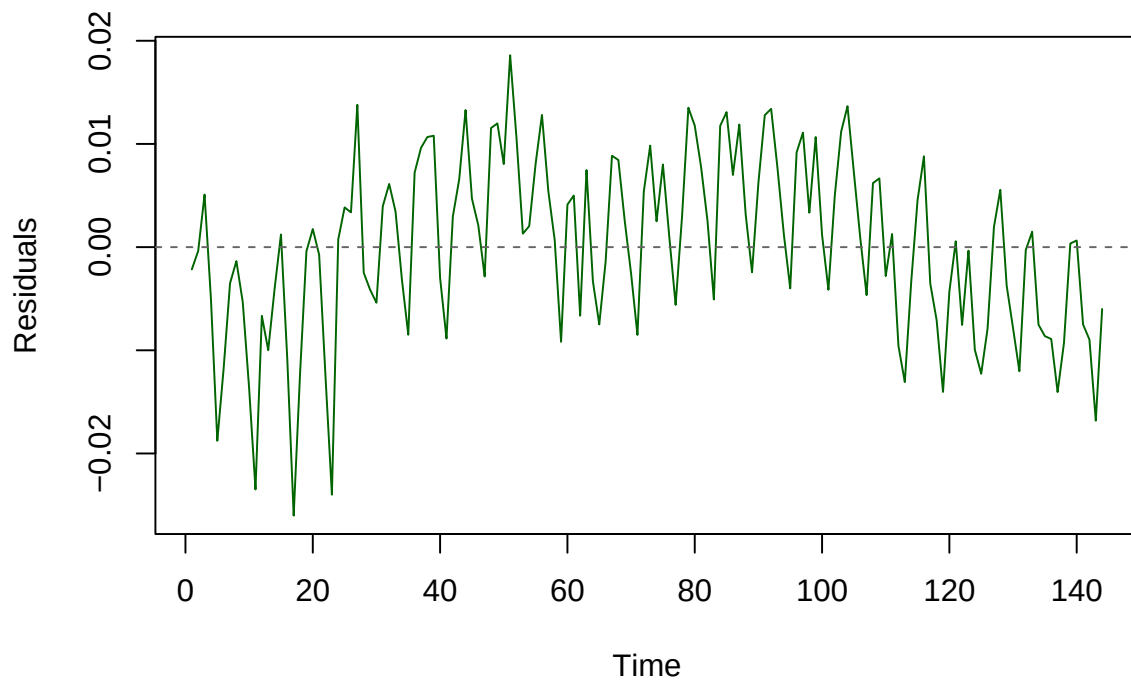


```

plot(resid_bc_k1, type = "l", col = "darkgreen",
     main = "Residuals from Harmonic Regression (Box-Cox, k = 1)",
     ylab = "Residuals", xlab = "Time")
abline(h = 0, lty = "dashed", col = "gray40")

```

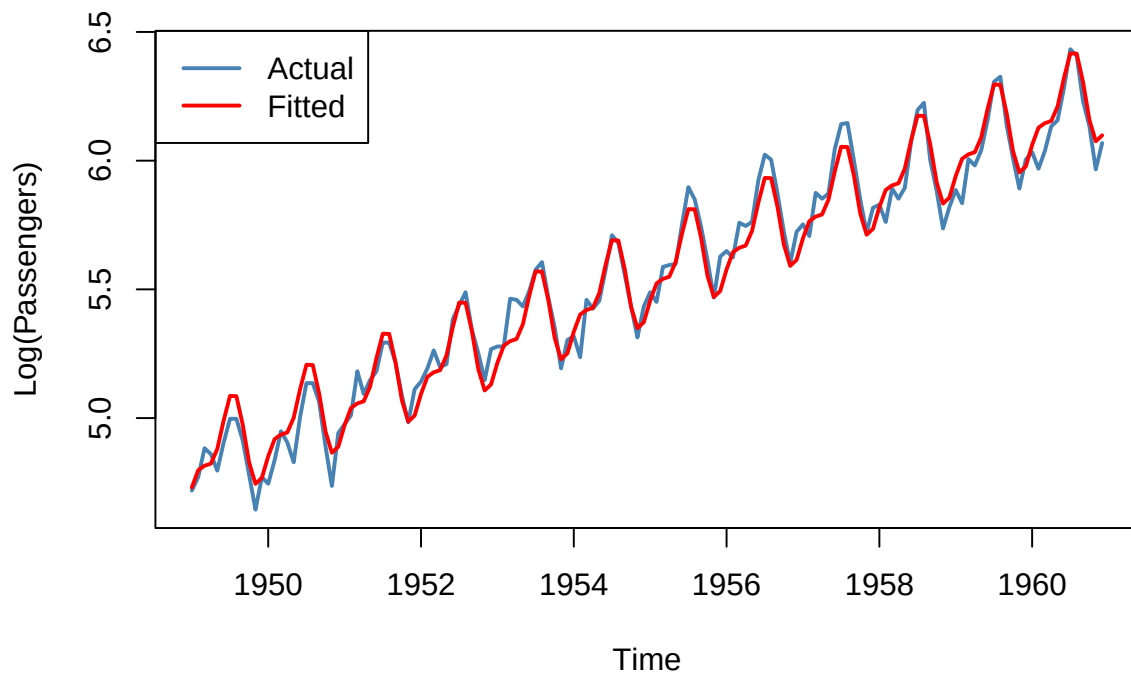
Residuals from Harmonic Regression (Box-Cox, k = 1)



```
time_vals_log_k2 <- time(log_ap_ts)
harm_model_log_k2 <- lm(log_ap_ts ~ time_vals_log_k2 + harmonic(log_ap_ts, 2))
fitted_log_k2 <- ts(fitted(harm_model_log_k2), start = start(log_ap_ts), frequency = frequency(log_ap_ts))
resid_log_k2 <- residuals(harm_model_log_k2)

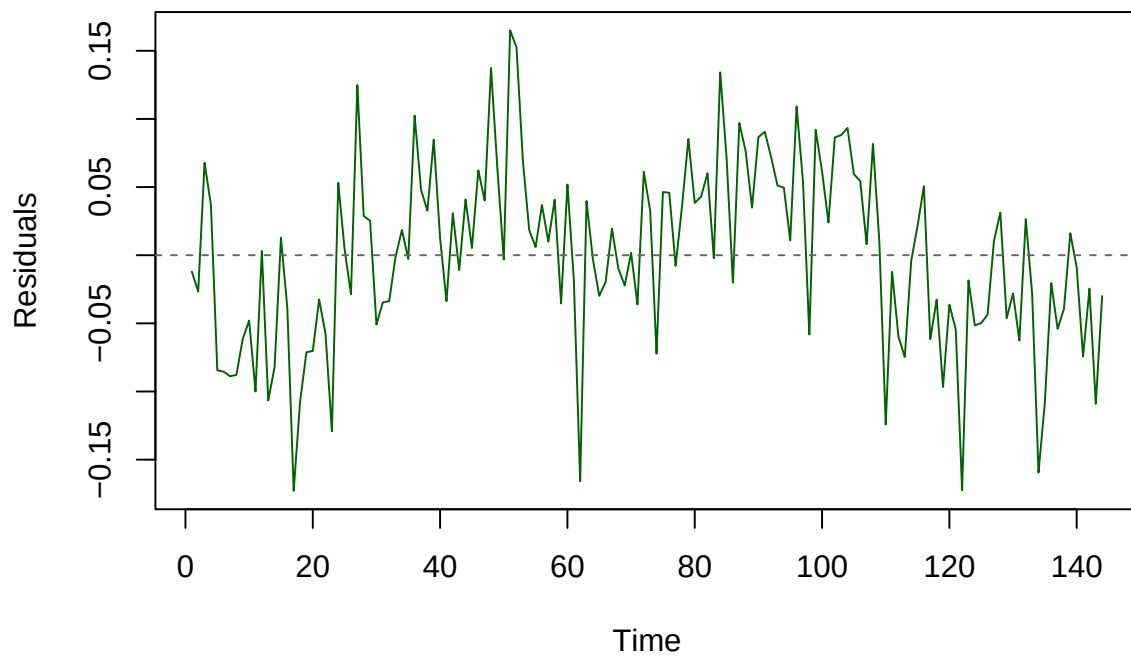
plot(log_ap_ts, type = "l", col = "steelblue", lwd = 2,
     ylab = "Log(Passengers)", xlab = "Time",
     main = "Log(AirPassengers) with Harmonic Fit (k = 2)")
lines(fitted_log_k2, col = "red", lwd = 2)
legend("topleft", legend = c("Actual", "Fitted"), col = c("steelblue", "red"), lwd = 2)
```

Log(AirPassengers) with Harmonic Fit (k = 2)



```
plot(resid_log_k2, type = "l", col = "darkgreen",  
     main = "Residuals from Harmonic Regression (Log, k = 2)",  
     ylab = "Residuals", xlab = "Time")  
abline(h = 0, lty = "dashed", col = "gray40")
```

Residuals from Harmonic Regression (Log, k = 2)



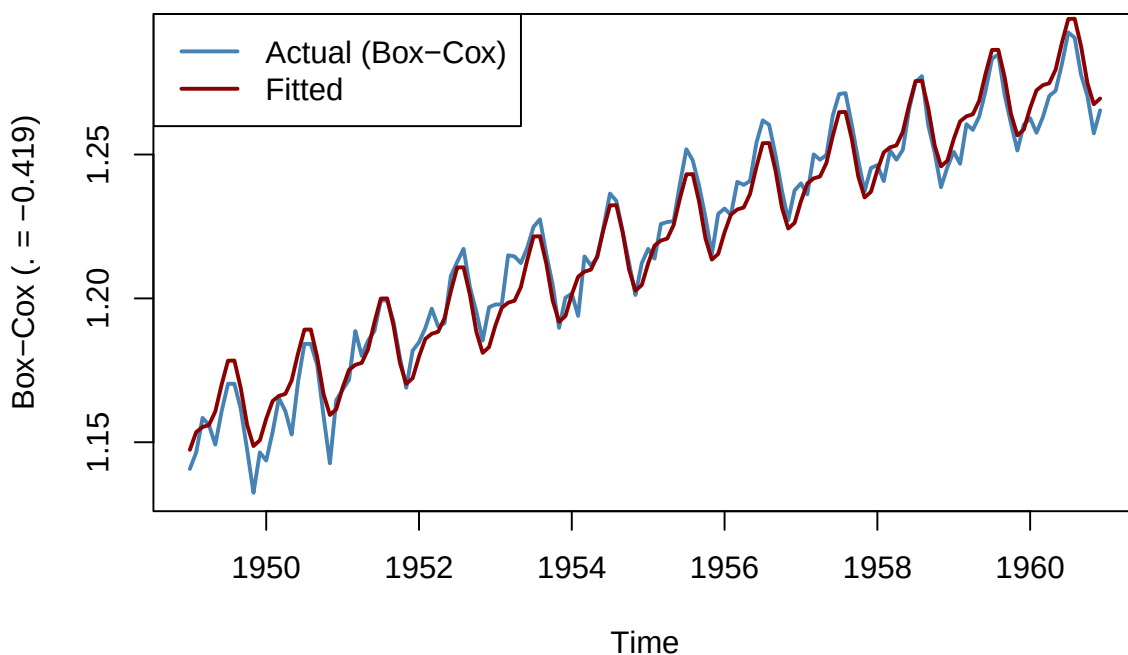
```

time_vals_bc_k2 <- time(ts_boxcox)
harm_model_bc_k2 <- lm(ts_boxcox ~ time_vals_bc_k2 + harmonic(ts_boxcox, 2))
fitted_bc_k2 <- ts(fitted(harm_model_bc_k2), start = start(ts_boxcox), frequency = frequency(ts_boxcox))
resid_bc_k2 <- residuals(harm_model_bc_k2)

plot(ts_boxcox, type = "l", col = "steelblue", lwd = 2,
     ylab = paste0("Box-Cox ( = ", round(lambda, 3), ")", xlab = "Time",
     main = paste0("Harmonic Regression with Box-Cox (k = 2)"))
lines(fitted_bc_k2, col = "darkred", lwd = 2)
legend("topleft", legend = c("Actual (Box-Cox)", "Fitted"), col = c("steelblue", "darkred"), lwd = 2)

```

Harmonic Regression with Box-Cox (k = 2)

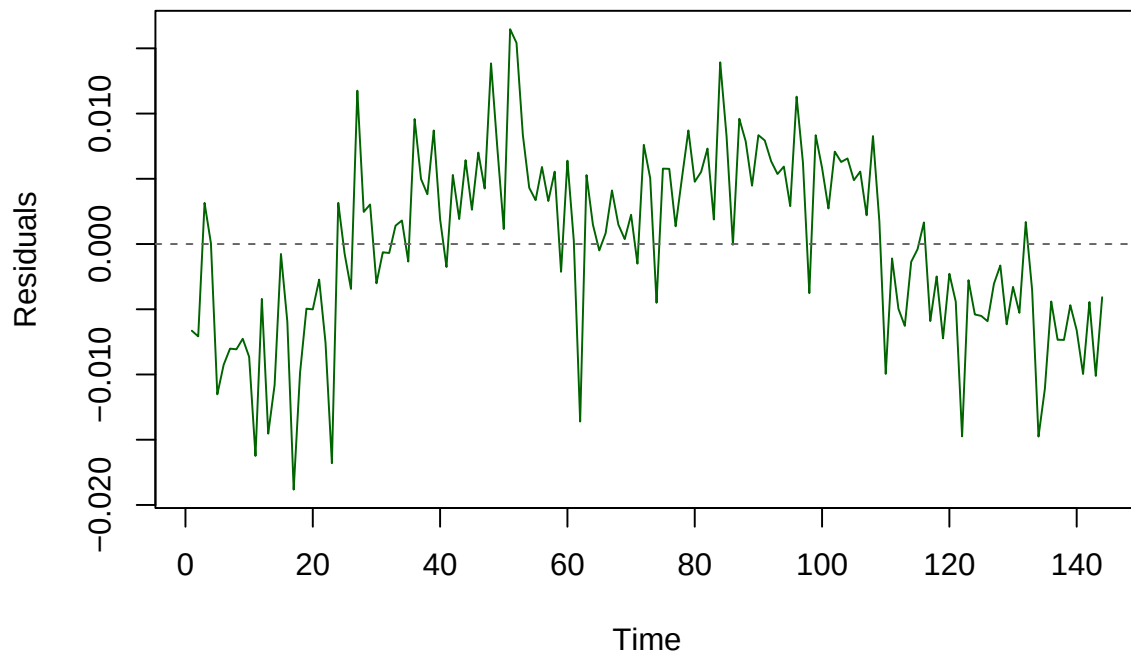


```

plot(resid_bc_k2, type = "l", col = "darkgreen",
     main = "Residuals from Harmonic Regression (Box-Cox, k = 2)",
     ylab = "Residuals", xlab = "Time")
abline(h = 0, lty = "dashed", col = "gray40")

```

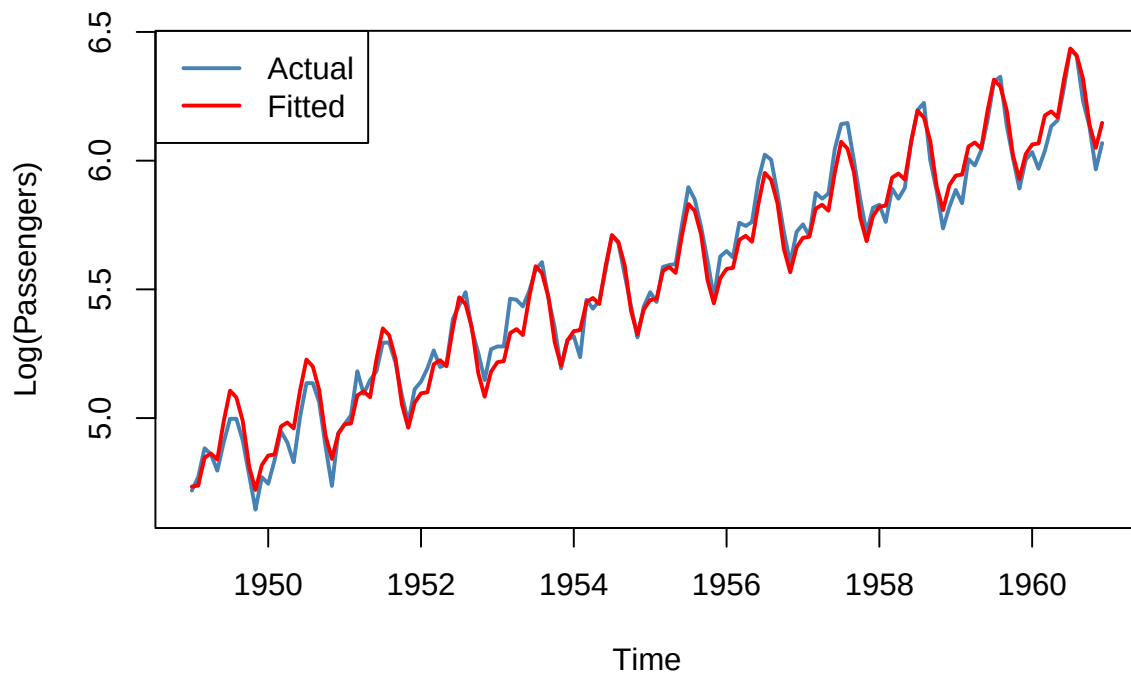
Residuals from Harmonic Regression (Box-Cox, $k = 2$)



```
time_vals_log_k4 <- time(log_ap_ts)
harm_model_log_k4 <- lm(log_ap_ts ~ time_vals_log_k4 + harmonic(log_ap_ts, 4))
fitted_log_k4 <- ts(fitted(harm_model_log_k4), start = start(log_ap_ts), frequency = frequency(log_ap_ts))
resid_log_k4 <- residuals(harm_model_log_k4)

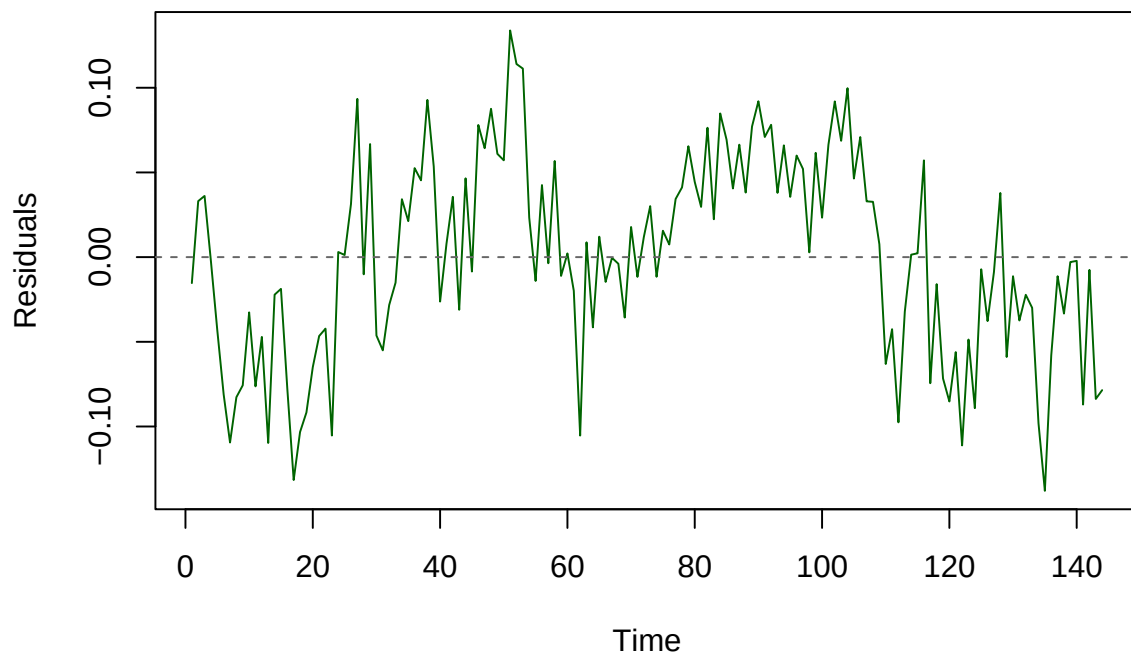
plot(log_ap_ts, type = "l", col = "steelblue", lwd = 2,
     ylab = "Log(Passengers)", xlab = "Time",
     main = "Log(AirPassengers) with Harmonic Fit (k = 4)")
lines(fitted_log_k4, col = "red", lwd = 2)
legend("topleft", legend = c("Actual", "Fitted"), col = c("steelblue", "red"), lwd = 2)
```

Log(AirPassengers) with Harmonic Fit (k = 4)



```
plot(resid_log_k4, type = "l", col = "darkgreen",  
     main = "Residuals from Harmonic Regression (Log, k = 4)",  
     ylab = "Residuals", xlab = "Time")  
abline(h = 0, lty = "dashed", col = "gray40")
```

Residuals from Harmonic Regression (Log, k = 4)



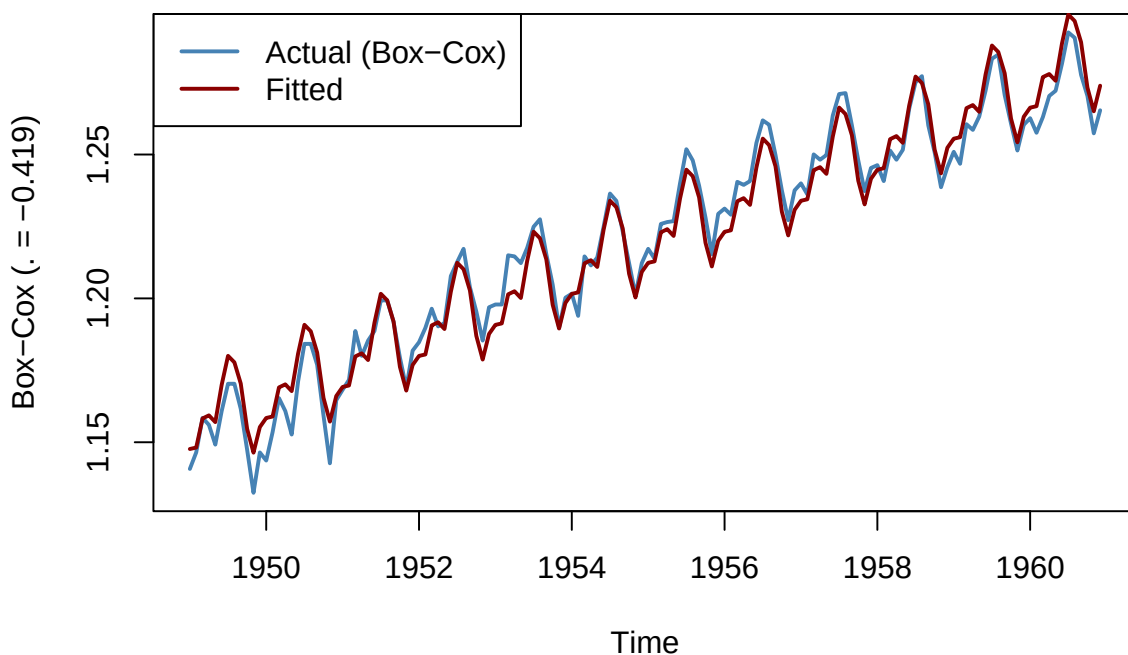
```

time_vals_bc_k4 <- time(ts_boxcox)
harm_model_bc_k4 <- lm(ts_boxcox ~ time_vals_bc_k4 + harmonic(ts_boxcox, 4))
fitted_bc_k4 <- ts(fitted(harm_model_bc_k4), start = start(ts_boxcox), frequency = frequency(ts_boxcox))
resid_bc_k4 <- residuals(harm_model_bc_k4)

plot(ts_boxcox, type = "l", col = "steelblue", lwd = 2,
     ylab = paste0("Box-Cox ( = ", round(lambda, 3), " )"), xlab = "Time",
     main = paste0("Harmonic Regression with Box-Cox (k = 4)"))
lines(fitted_bc_k4, col = "darkred", lwd = 2)
legend("topleft", legend = c("Actual (Box-Cox)", "Fitted"), col = c("steelblue", "darkred"), lwd = 2)

```

Harmonic Regression with Box-Cox (k = 4)

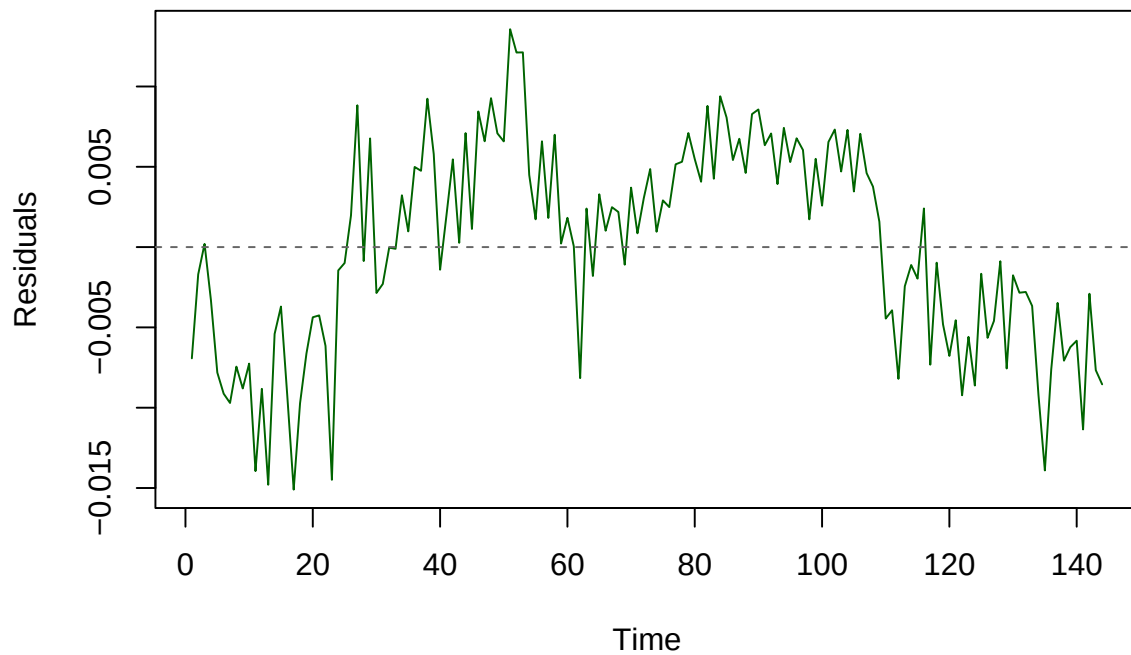


```

plot(resid_bc_k4, type = "l", col = "darkgreen",
     main = "Residuals from Harmonic Regression (Box-Cox, k = 4)",
     ylab = "Residuals", xlab = "Time")
abline(h = 0, lty = "dashed", col = "gray40")

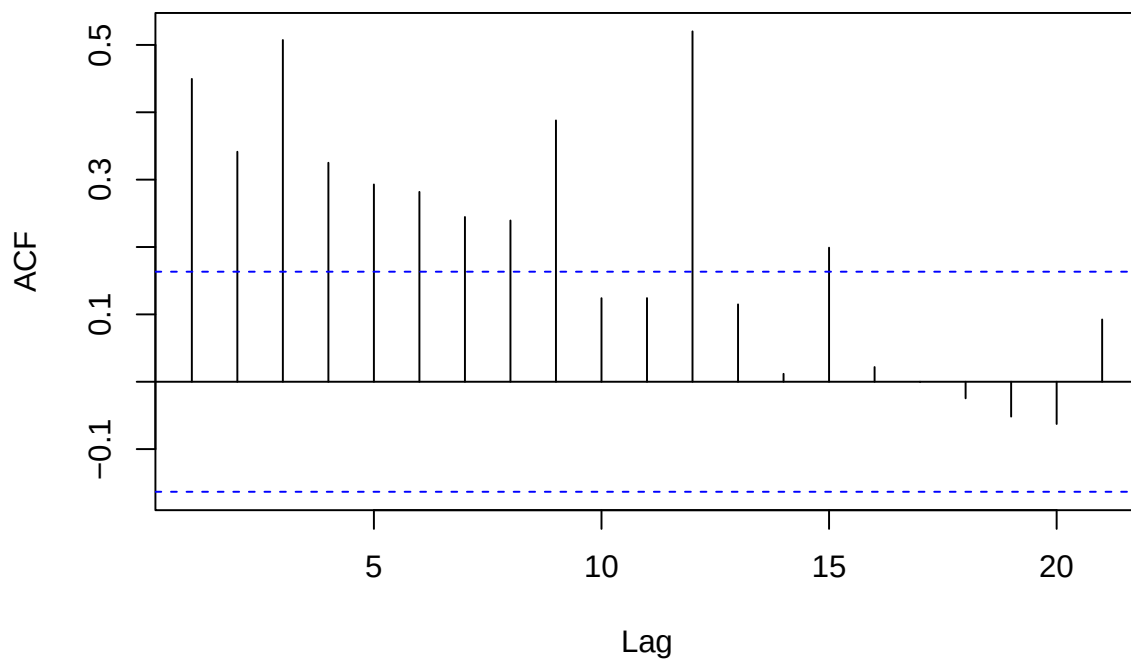
```

Residuals from Harmonic Regression (Box-Cox, $k = 4$)



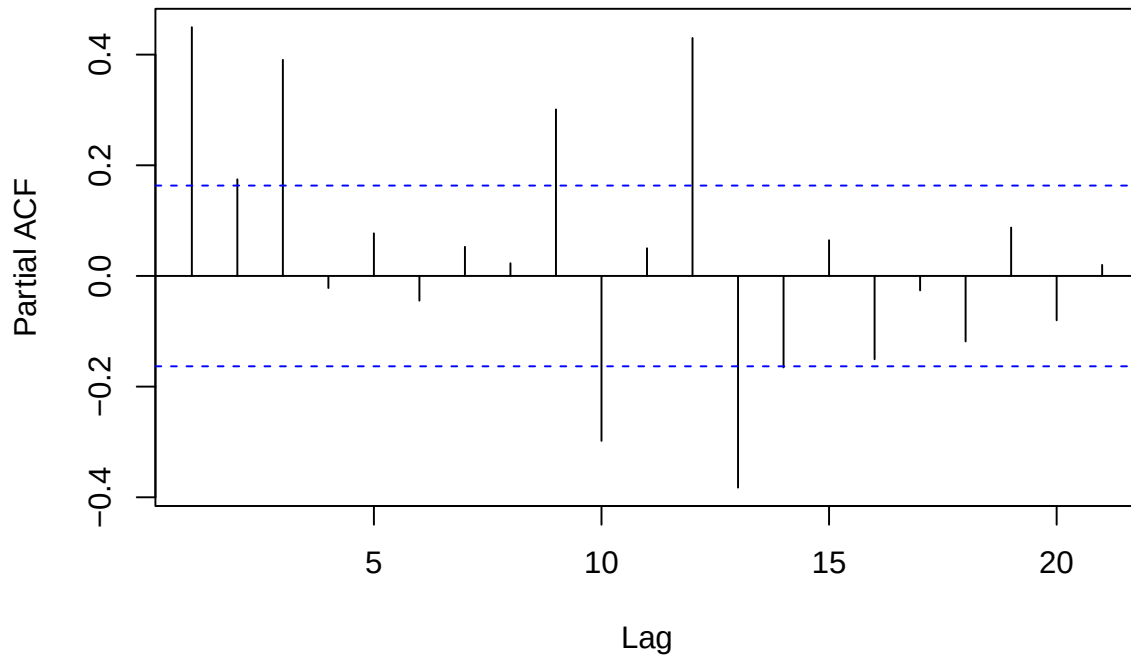
```
# ACF and PACF of residuals (log model, k = 2)
acf(resid_log_k2, main = "ACF of Harmonic Regression Residuals (Log, k = 2)")
```

ACF of Harmonic Regression Residuals (Log, $k = 2$)



```
pacf(resid_log_k2, main = "PACF of Harmonic Regression Residuals (Log, k = 2)")
```

PACF of Harmonic Regression Residuals (Log, k = 2)



```
tseries::adf.test(resid_log_k2)
```

```
##
## Augmented Dickey-Fuller Test
##
## data: resid_log_k2
## Dickey-Fuller = -2.6524, Lag order = 5, p-value = 0.3048
## alternative hypothesis: stationary
```

```
library(randtests)
```

```
##
## Attaching package: 'randtests'
##
## The following object is masked from 'package:tseries':
##
## runs.test
```

```
Box.test(resid_log_k2, lag = 20, type = "Ljung-Box")
```

```
##
## Box-Ljung test
##
## data: resid_log_k2
## X-squared = 225.56, df = 20, p-value < 2.2e-16
```

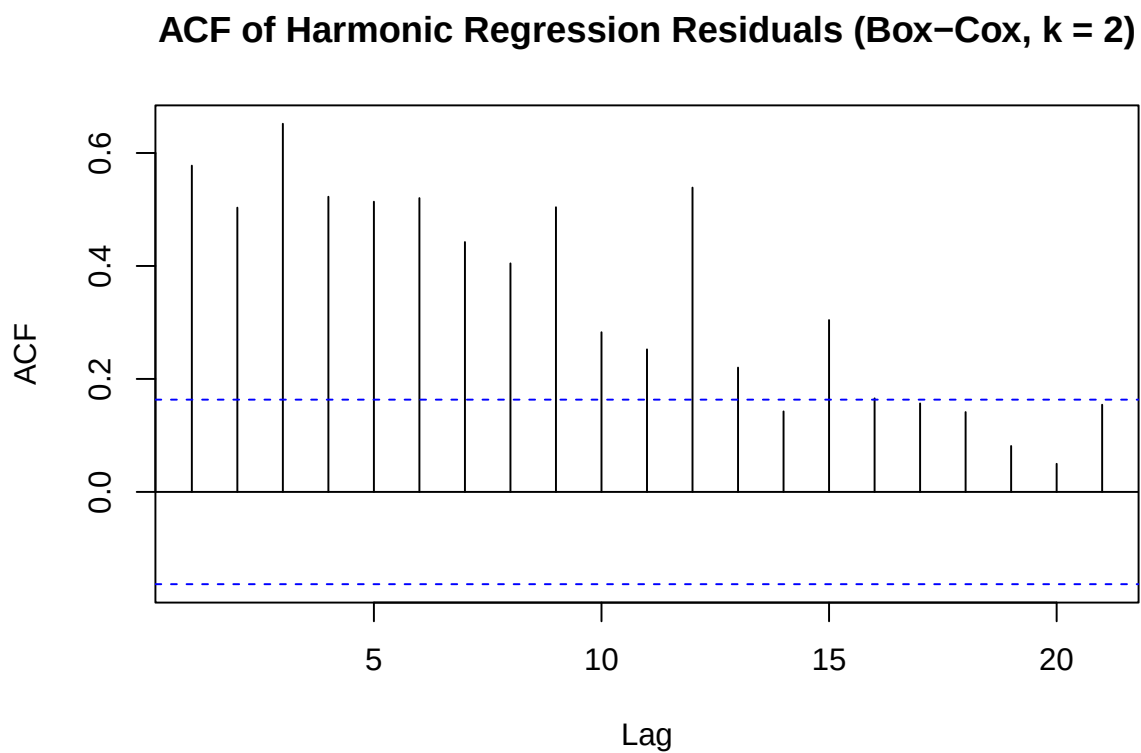
```
runs(resid_log_k2)
```

```
## $pvalue
## [1] 7.38e-05
##
## $observed.runs
```

```
## [1] 49
##
## $expected.runs
## [1] 72.98611
##
## $n1
## [1] 71
##
## $n2
## [1] 73
##
## $k
## [1] 0
```

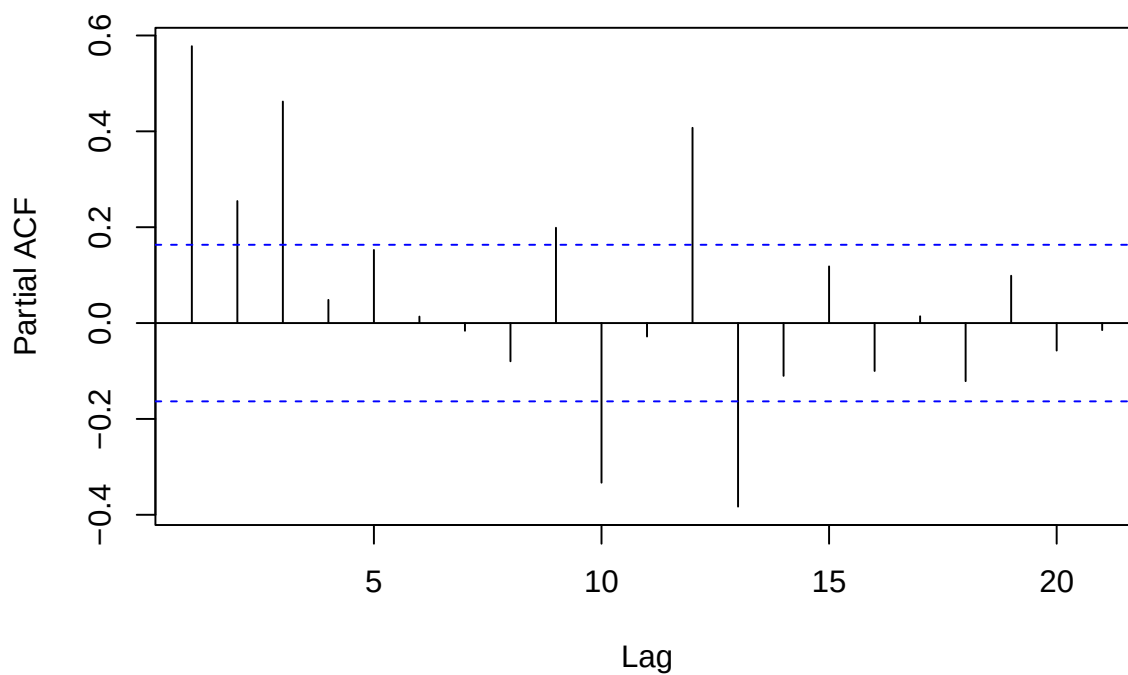
```
# ACF and PACF plots for Box-Cox residuals
```

```
acf(resid_bc_k2, main = "ACF of Harmonic Regression Residuals (Box-Cox, k = 2)")
```



```
pacf(resid_bc_k2, main = "PACF of Harmonic Regression Residuals (Box-Cox, k = 2)")
```

PACF of Harmonic Regression Residuals (Box-Cox, $k = 2$)



```
# Augmented Dickey-Fuller test  
tseries::adf.test(resid_bc_k2)
```

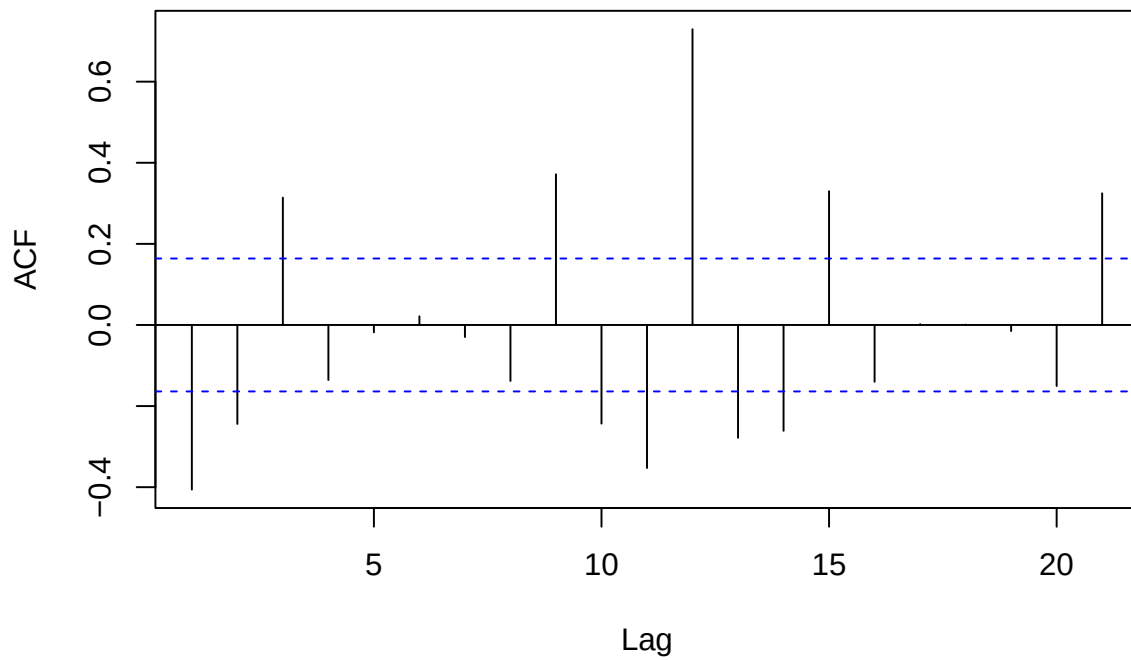
```
##  
## Augmented Dickey-Fuller Test  
##  
## data: resid_bc_k2  
## Dickey-Fuller = -1.6293, Lag order = 5, p-value = 0.7308  
## alternative hypothesis: stationary
```

```
Box.test(resid_bc_k2, lag = 20, type = "Ljung-Box")
```

```
##  
## Box-Ljung test  
##  
## data: resid_bc_k2  
## X-squared = 475.27, df = 20, p-value < 2.2e-16
```

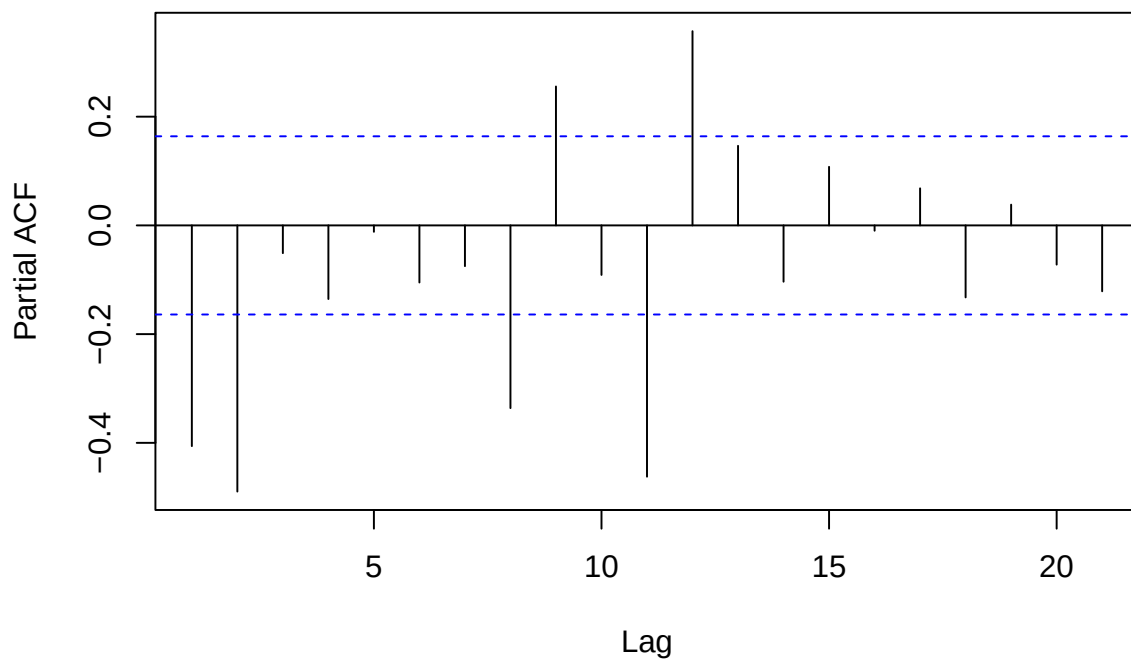
```
diff_resid <- diff(resid_log_k2)  
acf(diff_resid)
```

Series diff_resid



```
pacf(diff_resid)
```

Series diff_resid



```
tseries::adf.test(diff_resid)
```

```
## Warning in tseries::adf.test(diff_resid): p-value smaller than printed p-value
```

```

##
## Augmented Dickey-Fuller Test
##
## data: diff_resid
## Dickey-Fuller = -6.6956, Lag order = 5, p-value = 0.01
## alternative hypothesis: stationary
Box.test(diff_resid, lag = 20, type = "Ljung")

##
## Box-Ljung test
##
## data: diff_resid
## X-squared = 235.68, df = 20, p-value < 2.2e-16
library(tseries)
kpss.test(diff_resid)

## Warning in kpss.test(diff_resid): p-value greater than printed p-value
##
## KPSS Test for Level Stationarity
##
## data: diff_resid
## KPSS Level = 0.027651, Truncation lag parameter = 4, p-value = 0.1
runs(diff_resid)

## $pvalue
## [1] 0.00891
##
## $observed.runs
## [1] 88
##
## $expected.runs
## [1] 72.07692
##
## $n1
## [1] 77
##
## $n2
## [1] 66
##
## $k
## [1] 0
library(TSA)
eacf(diff_resid)

## AR/MA
##   0 1 2 3 4 5 6 7 8 9 10 11 12 13
## 0 x x x o o o o o x x x x x x
## 1 x x x o o o o o x o x x x x
## 2 o x o o o o o o o o o x o x
## 3 x x o o o o o o o o o x o o
## 4 o x o o o o o o o o o x x o
## 5 o x o o o x o o o o o x x o

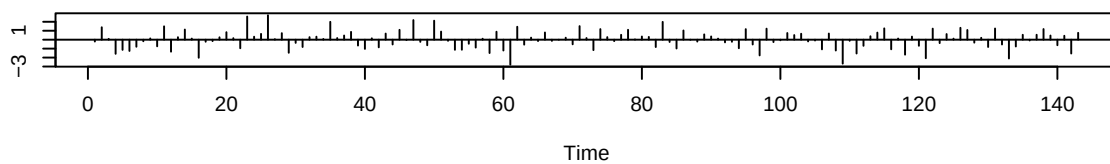
```

```
## 6 x x o x o x o o o o o x o o
## 7 x x o x o x o o o o o x o o
```

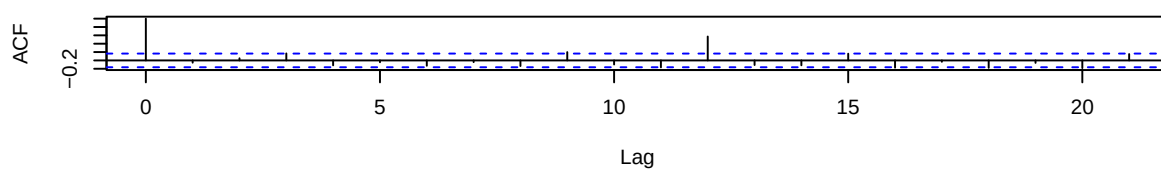
```
models <- list(
  arma_013 = arima(diff_resid, order = c(0, 0, 3)),
  arma_113 = arima(diff_resid, order = c(1, 0, 3)),
  arma_213 = arima(diff_resid, order = c(2, 0, 3)),
  arma_212 = arima(diff_resid, order = c(2, 0, 2))
)
```

```
tsdiag(models$arma_013)
```

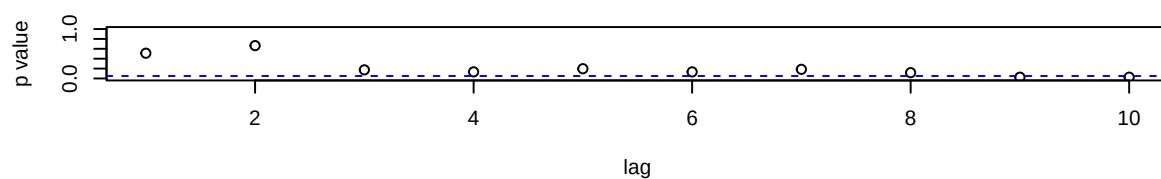
Standardized Residuals



ACF of Residuals

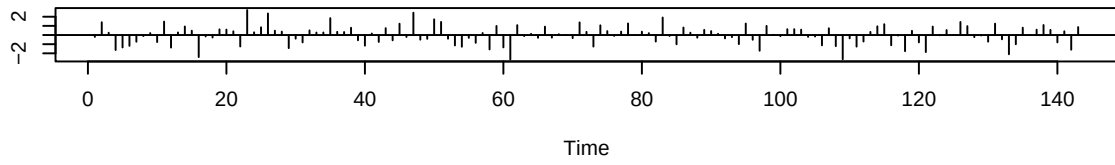


p values for Ljung-Box statistic

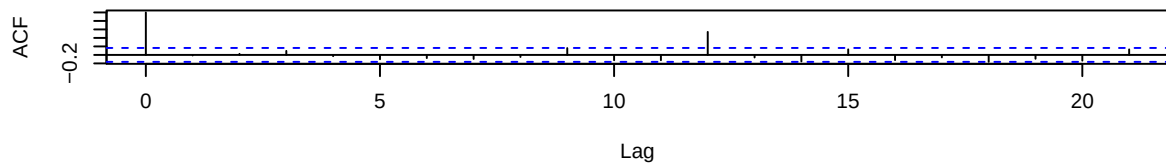


```
tsdiag(models$arma_113)
```

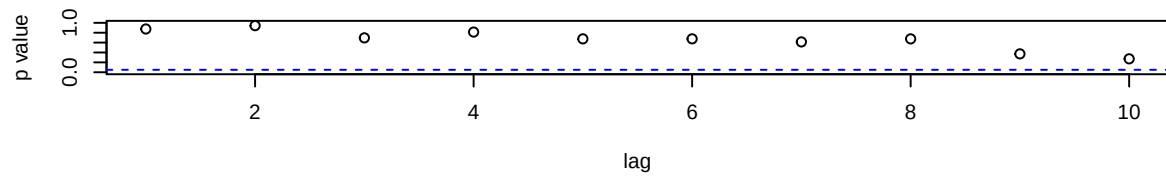
Standardized Residuals



ACF of Residuals

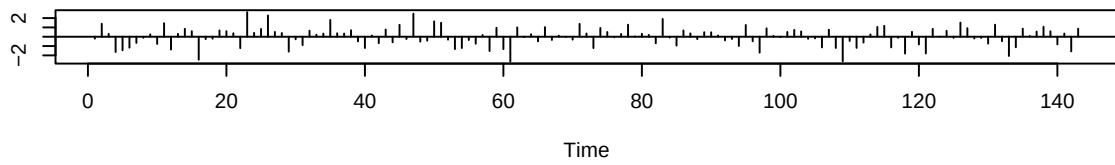


p values for Ljung-Box statistic

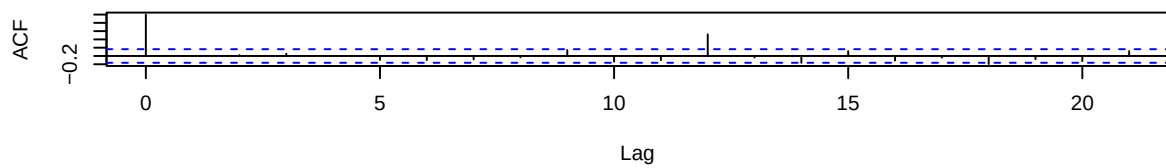


```
tsdiag(models$arma_213)
```

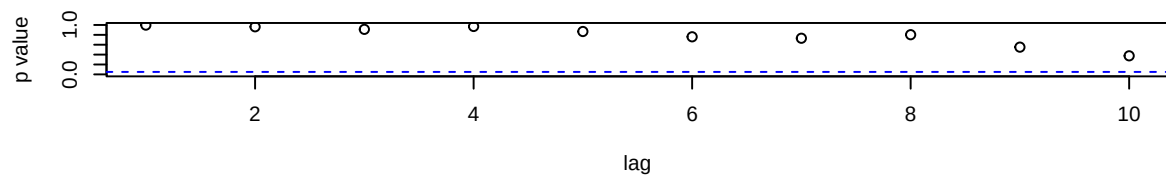
Standardized Residuals



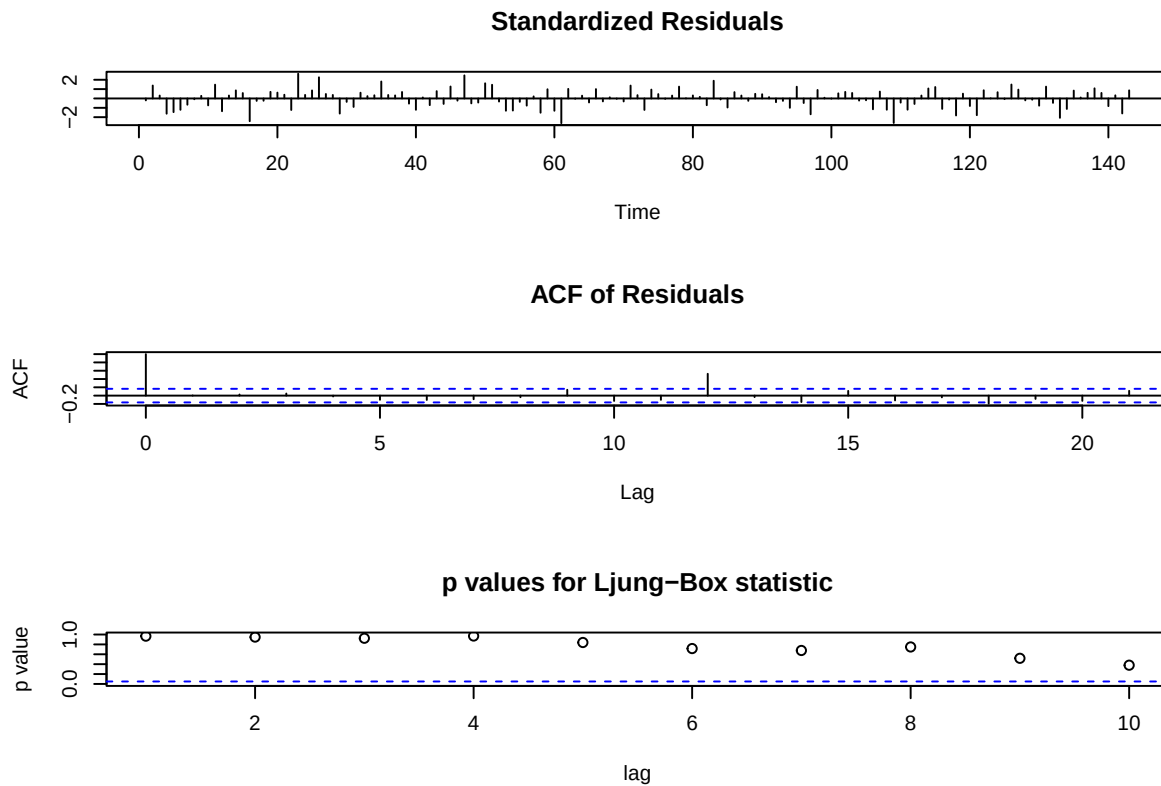
ACF of Residuals



p values for Ljung-Box statistic



```
tsdiag(models$arma_212)
```



```
# Load required package
library(tseries)

# Residuals
res_013 <- residuals(models$arma_013)
res_113 <- residuals(models$arma_113)
res_213 <- residuals(models$arma_213)
res_212 <- residuals(models$arma_212)

# --- ARMA(0,3) ---
aic_013 <- AIC(models$arma_013)
bic_013 <- BIC(models$arma_013)
lb_013 <- Box.test(res_013, lag = 20, type = "Ljung-Box")
runs_013 <- runs.test(res_013)
shapiro_013 <- shapiro.test(res_013)

# --- ARMA(1,3) ---
aic_113 <- AIC(models$arma_113)
bic_113 <- BIC(models$arma_113)
lb_113 <- Box.test(res_113, lag = 20, type = "Ljung-Box")
runs_113 <- runs.test(res_113)
shapiro_113 <- shapiro.test(res_113)

# --- ARMA(2,3) ---
aic_213 <- AIC(models$arma_213)
bic_213 <- BIC(models$arma_213)
```

```

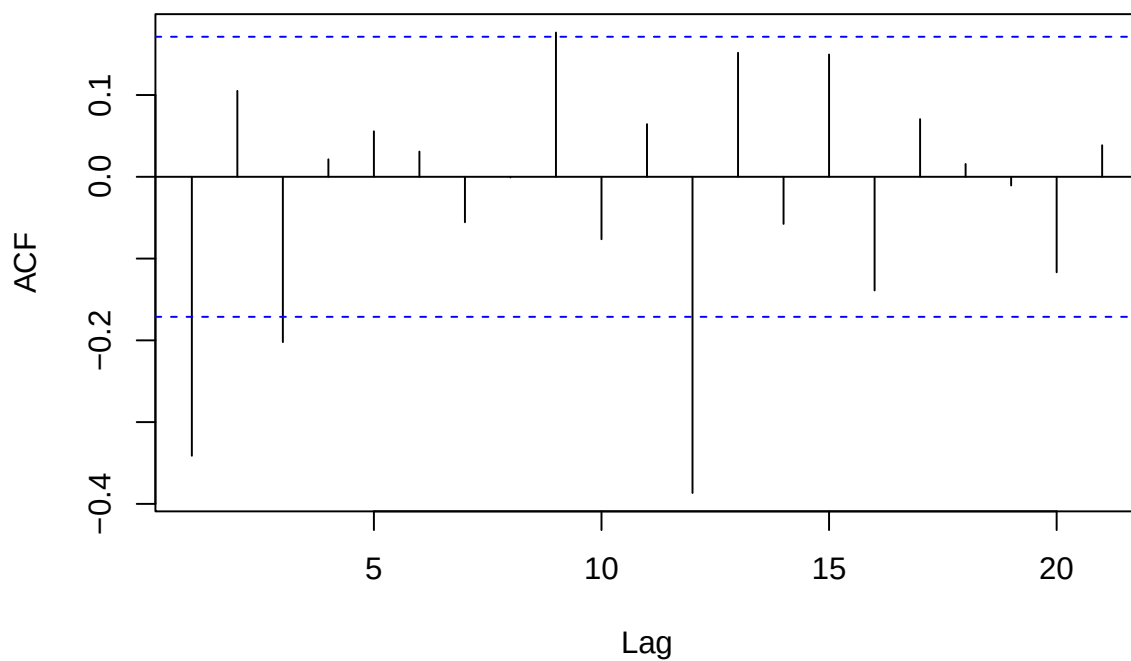
lb_213 <- Box.test(res_213, lag = 20, type = "Ljung-Box")
runs_213 <- runs.test(res_213)
shapiro_213 <- shapiro.test(res_213)

# --- ARMA(2,2) ---
aic_212 <- AIC(models$arma_212)
bic_212 <- BIC(models$arma_212)
lb_212 <- Box.test(res_212, lag = 20, type = "Ljung-Box")
runs_212 <- runs.test(res_212)
shapiro_212 <- shapiro.test(res_212)

final_diff<-diff(diff(resid_log_k2), lag = 12)
acf(final_diff)

```

Series final_diff

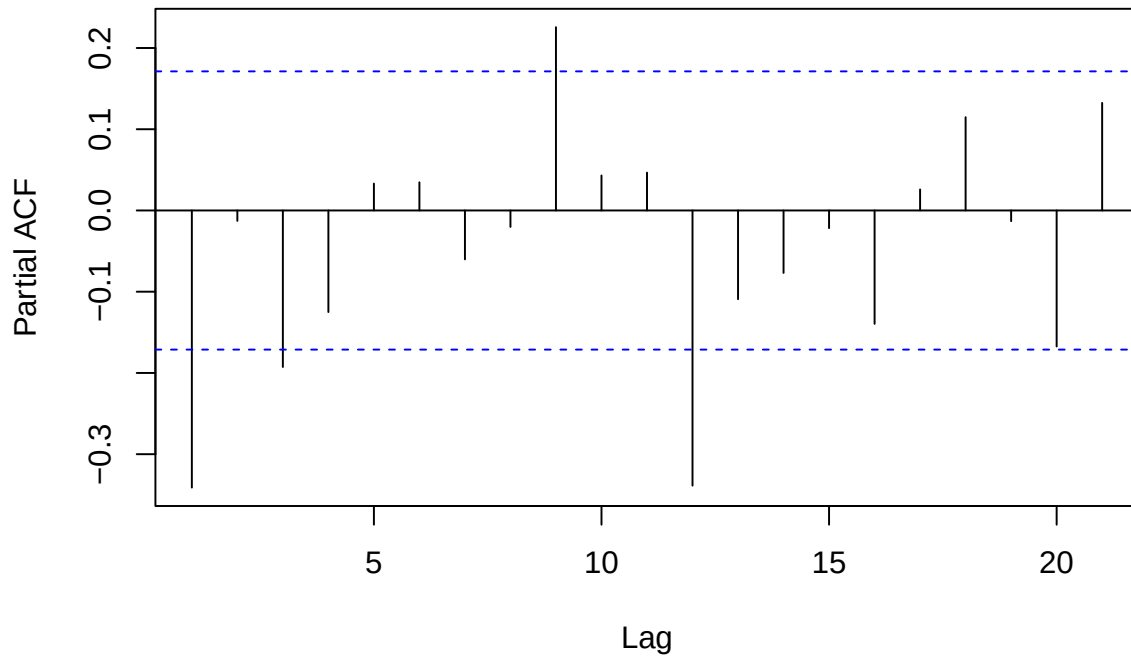


```

pacf(final_diff)

```

Series final_diff



```
tseries::adf.test(final_diff)
```

```
## Warning in tseries::adf.test(final_diff): p-value smaller than printed p-value
```

```
##
```

```
## Augmented Dickey-Fuller Test
```

```
##
```

```
## data: final_diff
```

```
## Dickey-Fuller = -5.1993, Lag order = 5, p-value = 0.01
```

```
## alternative hypothesis: stationary
```

```
Box.test(final_diff, lag = 20, type = "Ljung")
```

```
##
```

```
## Box-Ljung test
```

```
##
```

```
## data: final_diff
```

```
## X-squared = 64.598, df = 20, p-value = 1.352e-06
```

```
eacf(final_diff)
```

```
## AR/MA
```

```
## 0 1 2 3 4 5 6 7 8 9 10 11 12 13
```

```
## 0 x o x o o o o o o o o x o o
```

```
## 1 o o x o o o o o x o o x o o
```

```
## 2 o x x o o o o o x o o x o o
```

```
## 3 x x o x o o o o o o o x o o
```

```
## 4 x x x x o o o o o o o x o o
```

```
## 5 x o x o o o o o o o o x o o
```

```
## 6 x x o o o o o o o o o x o o
```

```
## 7 x x o o o o o o o o o x o o
```

```

# Fit SARIMA models
sarima_011_011 <- arima(resid_log_k2, order = c(0, 1, 1), seasonal = list(order = c(0, 1, 1), period =
sarima_111_011 <- arima(resid_log_k2, order = c(1, 1, 1), seasonal = list(order = c(0, 1, 1), period =
sarima_211_011 <- arima(resid_log_k2, order = c(2, 1, 1), seasonal = list(order = c(0, 1, 1), period =
sarima_112_011 <- arima(resid_log_k2, order = c(1, 1, 2), seasonal = list(order = c(0, 1, 1), period =
sarima_013_011 <- arima(resid_log_k2, order = c(0, 1, 3), seasonal = list(order = c(0, 1, 1), period =
sarima_113_011 <- arima(resid_log_k2, order = c(1, 1, 3), seasonal = list(order = c(0, 1, 1), period =

cat("AIC / BIC:\n")

## AIC / BIC:
cat("SARIMA(0,1,1):", AIC(sarima_011_011), "/", BIC(sarima_011_011), "\n")

## SARIMA(0,1,1): -483.393 / -474.7675
cat("SARIMA(1,1,1):", AIC(sarima_111_011), "/", BIC(sarima_111_011), "\n")

## SARIMA(1,1,1): -481.8931 / -470.3923
cat("SARIMA(2,1,1):", AIC(sarima_211_011), "/", BIC(sarima_211_011), "\n")

## SARIMA(2,1,1): -482.264 / -467.888
cat("SARIMA(1,1,2):", AIC(sarima_112_011), "/", BIC(sarima_112_011), "\n")

## SARIMA(1,1,2): -482.0359 / -467.6599
cat("SARIMA(0,1,3):", AIC(sarima_013_011), "/", BIC(sarima_013_011), "\n")

## SARIMA(0,1,3): -482.6368 / -468.2608
cat("SARIMA(1,1,3):", AIC(sarima_113_011), "/", BIC(sarima_113_011), "\n")

## SARIMA(1,1,3): -480.8145 / -463.5633
res011 <- residuals(sarima_011_011)
cat("SARIMA(0,1,1) Diagnostics:\n")

## SARIMA(0,1,1) Diagnostics:
print(Box.test(res011, lag = 20, type = "Ljung"))

##
## Box-Ljung test
##
## data: res011
## X-squared = 17.448, df = 20, p-value = 0.6237
print(shapiro.test(res011))

##
## Shapiro-Wilk normality test
##
## data: res011
## W = 0.98566, p-value = 0.1401
print(runs(res011))

## $pvalue
## [1] 0.145
##

```

```

## $observed.runs
## [1] 63
##
## $expected.runs
## [1] 72.11111
##
## $n1
## [1] 80
##
## $n2
## [1] 64
##
## $k
## [1] 0
res111 <- residuals(sarima_111_011)
cat("SARIMA(0,1,1) Diagnostics:\n")

## SARIMA(0,1,1) Diagnostics:
print(Box.test(res111, lag = 20, type = "Ljung"))

##
## Box-Ljung test
##
## data: res111
## X-squared = 16.457, df = 20, p-value = 0.6879
print(shapiro.test(res111))

##
## Shapiro-Wilk normality test
##
## data: res111
## W = 0.98276, p-value = 0.06742
print(runs(res111))

## $pvalue
## [1] 0.416
##
## $observed.runs
## [1] 67
##
## $expected.runs
## [1] 72.31944
##
## $n1
## [1] 79
##
## $n2
## [1] 65
##
## $k
## [1] 0

```

```

res211 <- residuals(sarima_211_011)
cat("SARIMA(0,1,1) Diagnostics:\n")

## SARIMA(0,1,1) Diagnostics:
print(Box.test(res211, lag = 20, type = "Ljung"))

##
## Box-Ljung test
##
## data: res211
## X-squared = 18.152, df = 20, p-value = 0.5774
print(shapiro.test(res211))

##
## Shapiro-Wilk normality test
##
## data: res211
## W = 0.97785, p-value = 0.01959
print(runs(res211))

## $pvalue
## [1] 0.4
##
## $observed.runs
## [1] 67
##
## $expected.runs
## [1] 72.5
##
## $n1
## [1] 78
##
## $n2
## [1] 66
##
## $k
## [1] 0

res112 <- residuals(sarima_112_011)
cat("SARIMA(0,1,1) Diagnostics:\n")

## SARIMA(0,1,1) Diagnostics:
print(Box.test(res112, lag = 20, type = "Ljung"))

##
## Box-Ljung test
##
## data: res112
## X-squared = 16.643, df = 20, p-value = 0.676
print(shapiro.test(res112))

##
## Shapiro-Wilk normality test

```

```

##
## data:  res112
## W = 0.97887, p-value = 0.02522
print(runs(res112))

## $pvalue
## [1] 0.435
##
## $observed.runs
## [1] 67
##
## $expected.runs
## [1] 72.11111
##
## $n1
## [1] 80
##
## $n2
## [1] 64
##
## $k
## [1] 0
res013 <- residuals(sarima_013_011)
cat("SARIMA(0,1,1) Diagnostics:\n")

## SARIMA(0,1,1) Diagnostics:
print(Box.test(res013, lag = 20, type = "Ljung"))

##
## Box-Ljung test
##
## data:  res013
## X-squared = 17.809, df = 20, p-value = 0.6
print(shapiro.test(res013))

##
## Shapiro-Wilk normality test
##
## data:  res013
## W = 0.98545, p-value = 0.1332
print(runs(res013))

## $pvalue
## [1] 0.386
##
## $observed.runs
## [1] 67
##
## $expected.runs
## [1] 72.65278
##
## $n1
## [1] 77

```

```

##
## $n2
## [1] 67
##
## $k
## [1] 0
res113 <- residuals(sarima_113_011)
cat("SARIMA(0,1,1) Diagnostics:\n")

## SARIMA(0,1,1) Diagnostics:
print(Box.test(res113, lag = 20, type = "Ljung"))

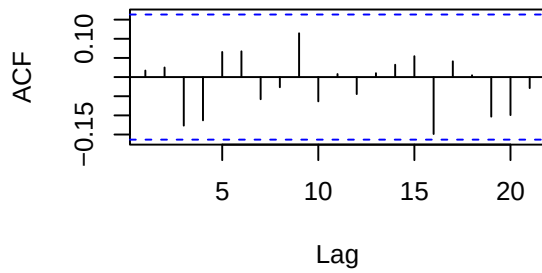
##
## Box-Ljung test
##
## data: res113
## X-squared = 17.884, df = 20, p-value = 0.595
print(shapiro.test(res113))

##
## Shapiro-Wilk normality test
##
## data: res113
## W = 0.98438, p-value = 0.1016
print(runs(res113))

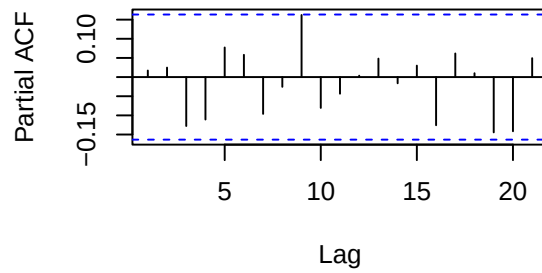
## $pvalue
## [1] 0.386
##
## $observed.runs
## [1] 67
##
## $expected.runs
## [1] 72.65278
##
## $n1
## [1] 77
##
## $n2
## [1] 67
##
## $k
## [1] 0
par(mfrow = c(2, 2))
acf(residuals(sarima_011_011), main = "ACF - SARIMA(0,1,1)")
pacf(residuals(sarima_011_011), main = "PACF - SARIMA(0,1,1)")
qqnorm(residuals(sarima_011_011), main = "QQ Plot - SARIMA(0,1,1)"); qqline(residuals(sarima_011_011), c
hist(residuals(sarima_011_011), col = "lightblue", main = "Histogram - SARIMA(0,1,1)", xlab = "Residual

```

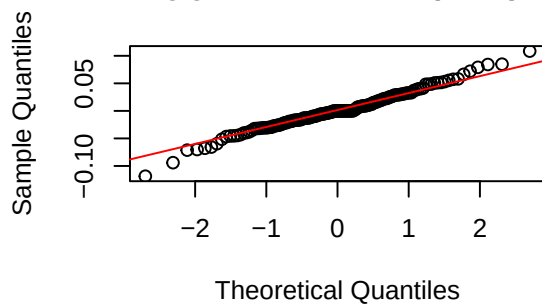
ACF – SARIMA(0,1,1)



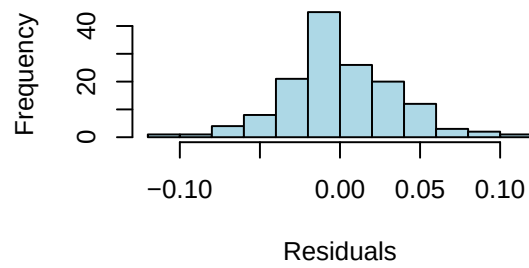
PACF – SARIMA(0,1,1)



QQ Plot – SARIMA(0,1,1)

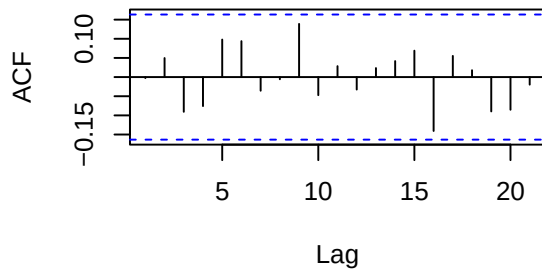


Histogram – SARIMA(0,1,1)

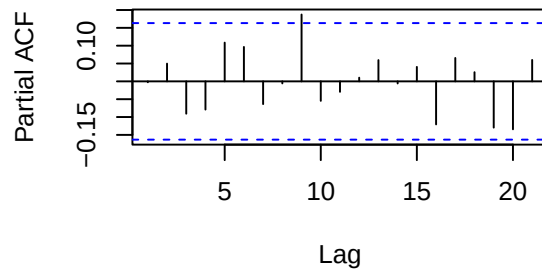


```
par(mfrow = c(2, 2))
acf(residuals(sarima_112_011), main = "ACF – SARIMA(1,1,1)")
pacf(residuals(sarima_112_011), main = "PACF – SARIMA(1,1,1)")
qqnorm(residuals(sarima_112_011), main = "QQ Plot – SARIMA(1,1,1)"); qqline(residuals(sarima_112_011), c)
hist(residuals(sarima_112_011), col = "lightblue", main = "Histogram – SARIMA(1,1,1)", xlab = "Residuals")
```

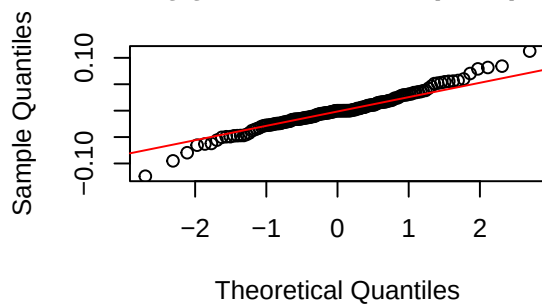
ACF – SARIMA(1,1,1)



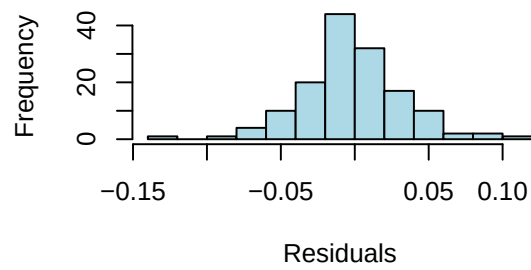
PACF – SARIMA(1,1,1)



QQ Plot – SARIMA(1,1,1)

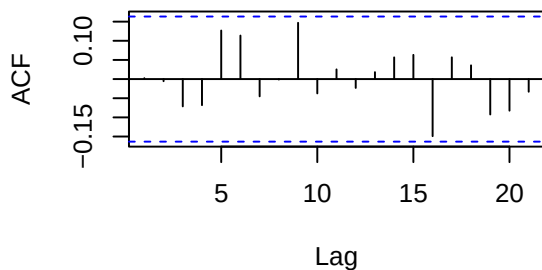


Histogram – SARIMA(1,1,1)

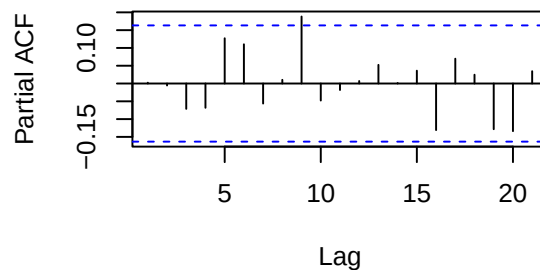


```
par(mfrow = c(2, 2))
acf(residuals(sarima_211_011), main = "ACF – SARIMA(2,1,1)")
pacf(residuals(sarima_211_011), main = "PACF – SARIMA(2,1,1)")
qqnorm(residuals(sarima_211_011), main = "QQ Plot – SARIMA(2,1,1)"); qqline(residuals(sarima_211_011), col = "red")
hist(residuals(sarima_211_011), col = "lightblue", main = "Histogram – SARIMA(2,1,1)", xlab = "Residuals")
```

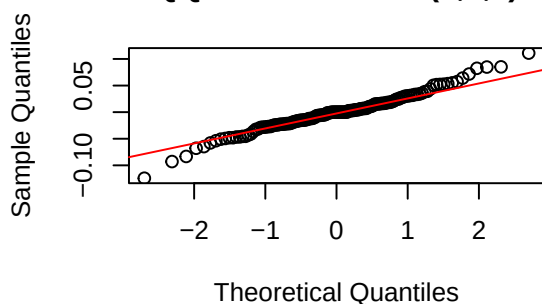
ACF – SARIMA(2,1,1)



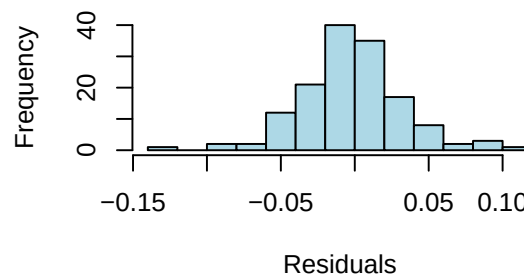
PACF – SARIMA(2,1,1)



QQ Plot – SARIMA(2,1,1)



Histogram – SARIMA(2,1,1)



```
library(forecast)

# Time series
resid_log_k2_ts <- ts(resid_log_k2, start = c(1949, 1), frequency = 12)
train_ts <- window(resid_log_k2_ts, end = c(1958, 12))
test_ts <- window(resid_log_k2_ts, start = c(1959, 1))

sarima_211 <- Arima(train_ts, order = c(2,1,1), seasonal = list(order = c(0,1,1), period = 12))
pred_211 <- predict(sarima_211, n.ahead = 24)

msfe_211 <- mean((test_ts - pred_211$pred)^2)
mafe_211 <- mean(abs(test_ts - pred_211$pred))

cat("SARIMA(2,1,1)(0,1,1)[12]\n")

## SARIMA(2,1,1)(0,1,1)[12]
cat("MSFE:", round(msfe_211, 5), "\n")

## MSFE: 0.00135
cat("MAFE:", round(mafe_211, 5), "\n\n")

## MAFE: 0.03182

sarima_112 <- Arima(train_ts, order = c(1,1,2), seasonal = list(order = c(0,1,1), period = 12))
pred_112 <- predict(sarima_112, n.ahead = 24)

msfe_112 <- mean((test_ts - pred_112$pred)^2)
mafe_112 <- mean(abs(test_ts - pred_112$pred))
```

```

cat("SARIMA(1,1,2)(0,1,1)[12]\n")

## SARIMA(1,1,2)(0,1,1)[12]
cat("MSFE:", round(msfe_112, 5), "\n")

## MSFE: 0.00645
cat("MAFE:", round(mafe_112, 5), "\n")

## MAFE: 0.07411
rolling_sarima <- function(y, order, seasonal, period = 12, h = 1, window_size = 120) {
  s <- length(y) - window_size + 1
  ehat <- numeric(s)

  for (j in 1:s) {
    train_sub <- y[j:(j + window_size - h - 1)]
    test_sub <- y[(j + window_size - h)]

    model <- Arima(train_sub, order = order,
                   seasonal = list(order = seasonal, period = period))

    forecast_val <- predict(model, n.ahead = h)$pred[h]
    ehat[j] <- test_sub - forecast_val
  }

  return(ehat)
}

resid_log_k2_ts <- ts(resid_log_k2, start = c(1949, 1), frequency = 12)

err_211 <- rolling_sarima(resid_log_k2_ts, c(0,1,1), c(0,1,1))
msfe_roll_211 <- mean(err_211^2)
mafe_roll_211 <- mean(abs(err_211))

err_112 <- rolling_sarima(resid_log_k2_ts, c(1,1,2), c(0,1,1))
msfe_roll_112 <- mean(err_112^2)
mafe_roll_112 <- mean(abs(err_112))

cat("Rolling Forecast Evaluation:\n")

## Rolling Forecast Evaluation:
cat("SARIMA(2,1,1): MSFE =", round(msfe_roll_211, 5), ", MAFE =", round(mafe_roll_211, 5), "\n")

## SARIMA(2,1,1): MSFE = 0.00122 , MAFE = 0.02647
cat("SARIMA(1,1,2): MSFE =", round(msfe_roll_112, 5), ", MAFE =", round(mafe_roll_112, 5), "\n")

## SARIMA(1,1,2): MSFE = 0.00127 , MAFE = 0.02696
sarima_011 <- Arima(train_ts, order = c(0,1,1), seasonal = list(order = c(0,1,1), period = 12))
pred_011 <- predict(sarima_011, n.ahead = 24)

```

```

e_211 <- test_ts - pred_211$pred
e_011 <- test_ts - pred_011$pred

# Diebold-Mariano test
d <- e_211^2 - e_011^2
dbar <- mean(d)
gamma0 <- acf(d, type = "covariance", plot = FALSE)$acf[1]
DM_stat <- dbar / sqrt(gamma0 / (length(d) - 1))

crit_val <- qt(0.975, df = length(d) - 1)

cat("DM Statistic:", round(DM_stat, 4), "\n")

## DM Statistic: -6.9389
cat("Critical Value:", round(crit_val, 4), "\n")

## Critical Value: 2.0687

library(forecast)
library(ggplot2)
library(zoo)

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##      as.Date, as.Date.numeric

resid_log_k2_ts <- ts(resid_log_k2, start = c(1949, 1), frequency = 12)

train_ts <- window(resid_log_k2_ts, end = c(1958, 12))
test_ts <- window(resid_log_k2_ts, start = c(1959, 1))

sarima_011 <- Arima(train_ts, order = c(0,1,1), seasonal = list(order = c(0,1,1), period = 12))
fcast_011 <- forecast(sarima_011, h = 24)

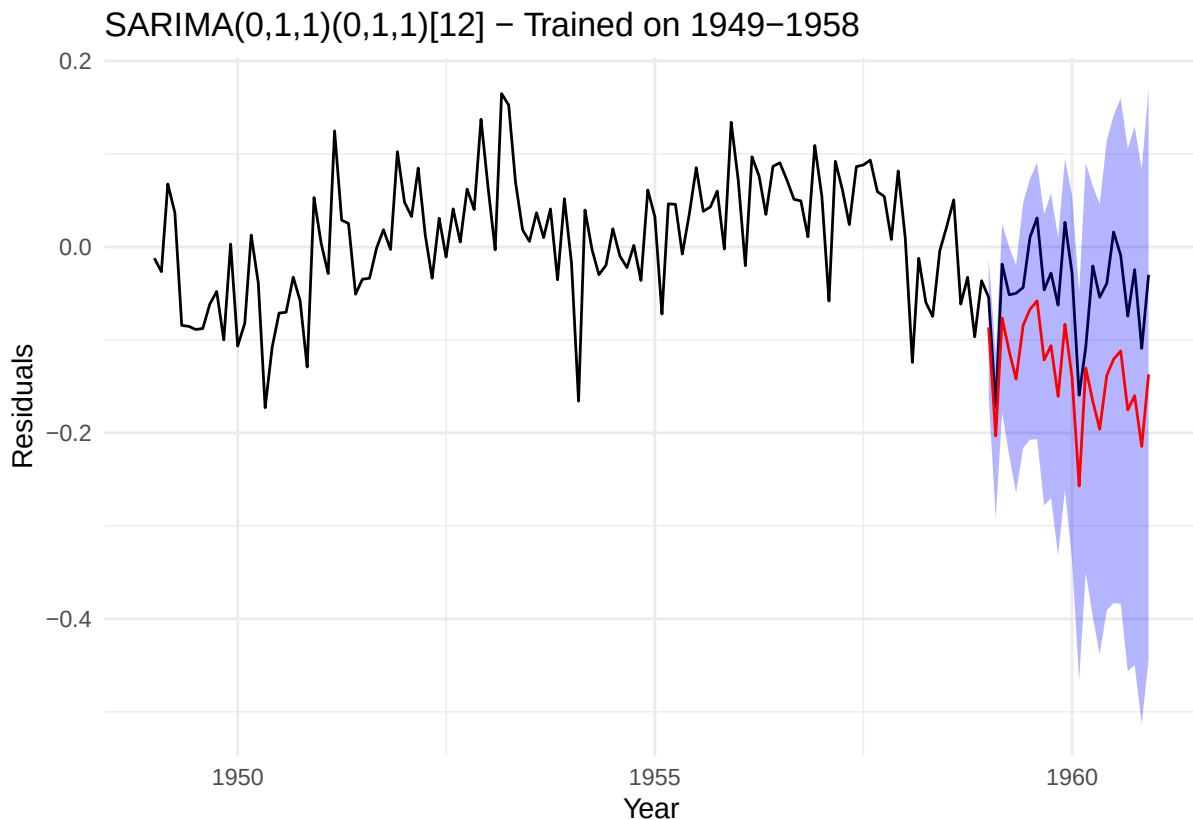
# Forecast data
fcast_df_011 <- data.frame(
  Date = as.Date(as.yearmon(time(fcast_011$mean))),
  Forecast = as.numeric(fcast_011$mean),
  Lower = as.numeric(fcast_011$lower[, 2]),
  Upper = as.numeric(fcast_011$upper[, 2])
)

# Full series for plotting
full_df <- data.frame(
  Date = as.Date(as.yearmon(time(resid_log_k2_ts))),
  Value = as.numeric(resid_log_k2_ts)
)

```

```
# Plot
ggplot() +
  geom_line(data = full_df, aes(x = Date, y = Value), color = "black") +
  geom_ribbon(data = fcast_df_011, aes(x = Date, ymin = Lower, ymax = Upper), fill = "blue", alpha = 0.5) +
  geom_line(data = fcast_df_011, aes(x = Date, y = Forecast), color = "red") +

  ggtitle("SARIMA(0,1,1)(0,1,1)[12] - Trained on 1949-1958") +
  ylab("Residuals") + xlab("Year") +
  theme_minimal()
```

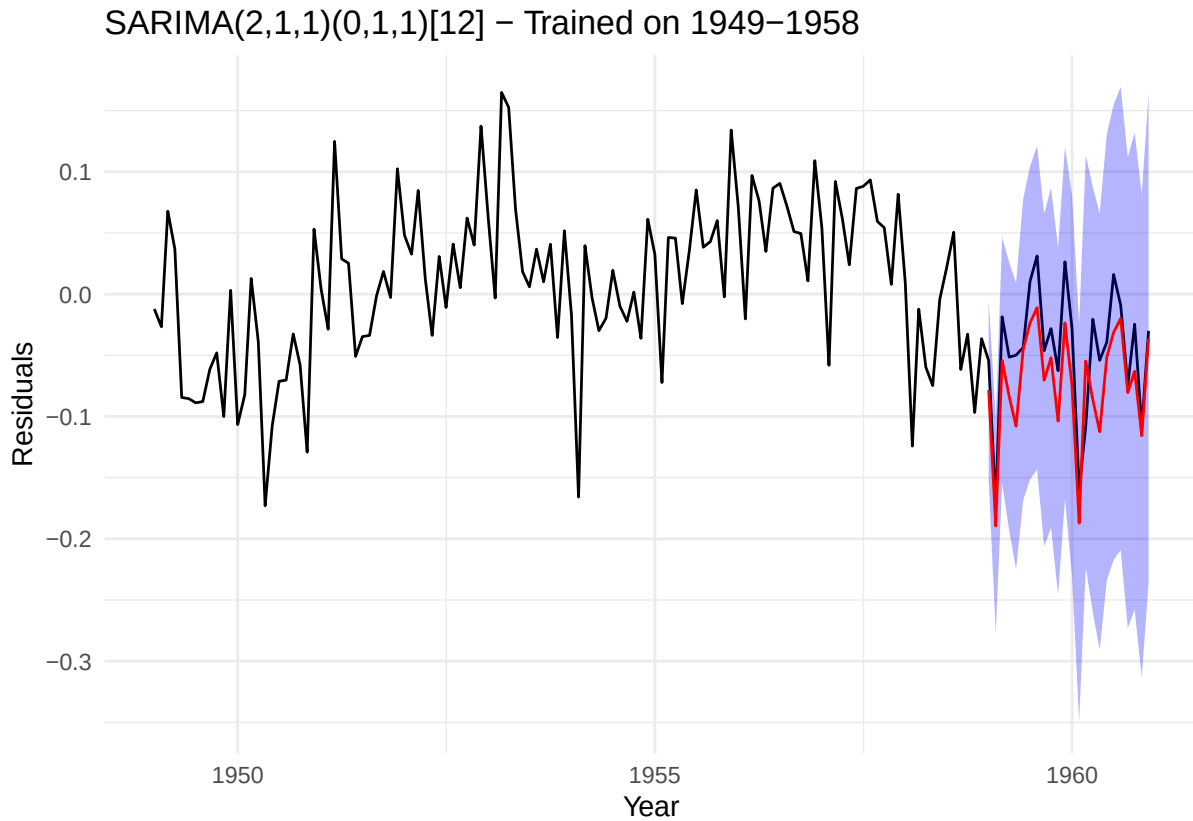


```
sarima_211 <- Arima(train_ts, order = c(2,1,1), seasonal = list(order = c(0,1,1), period = 12))
fcast_211 <- forecast(sarima_211, h = 24)

fcast_df_211 <- data.frame(
  Date = as.Date(as.yearmon(time(fcast_211$mean))),
  Forecast = as.numeric(fcast_211$mean),
  Lower = as.numeric(fcast_211$lower[, 2]),
  Upper = as.numeric(fcast_211$upper[, 2])
)

ggplot() +
  geom_line(data = full_df, aes(x = Date, y = Value), color = "black") +
  geom_ribbon(data = fcast_df_211, aes(x = Date, ymin = Lower, ymax = Upper), fill = "blue", alpha = 0.5) +
  geom_line(data = fcast_df_211, aes(x = Date, y = Forecast), color = "red") +
```

```
ggtitle("SARIMA(2,1,1)(0,1,1)[12] - Trained on 1949-1958") +
ylab("Residuals") + xlab("Year") +
theme_minimal()
```

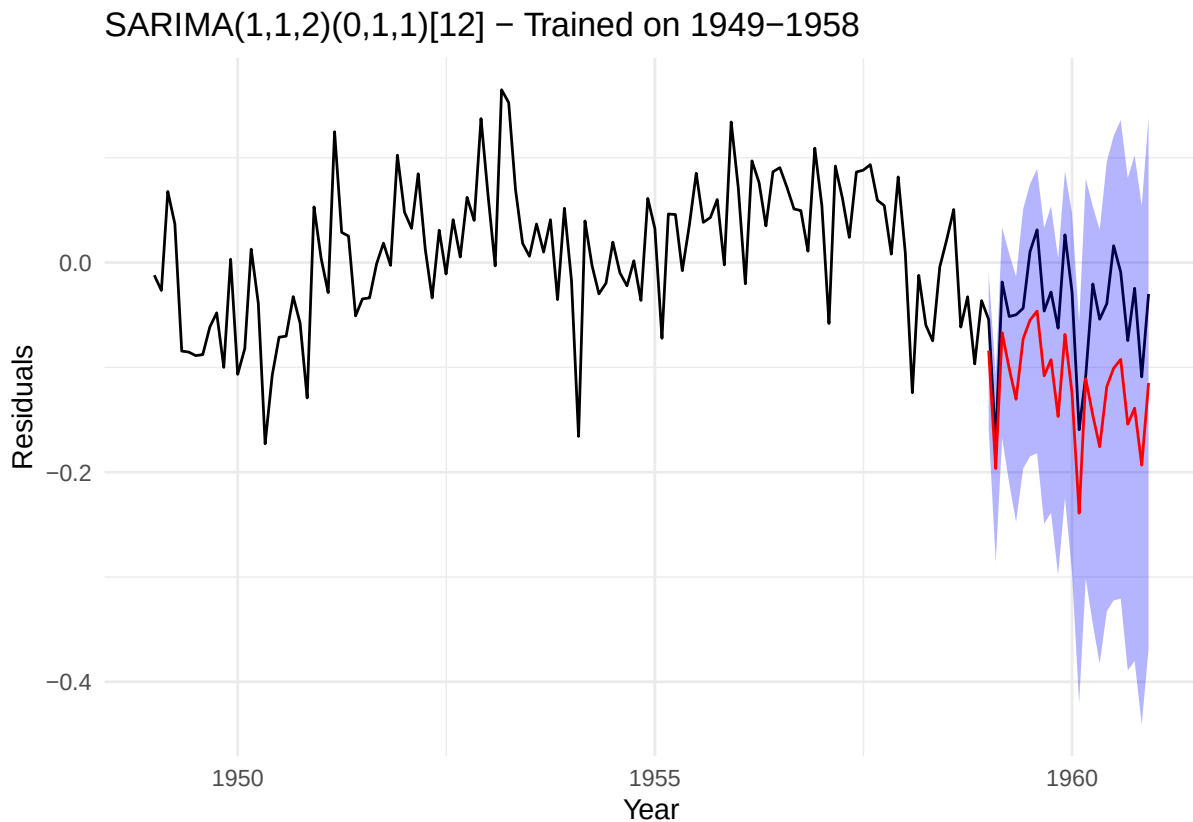


```
# Fit & forecast
sarima_112 <- Arima(train_ts, order = c(1,1,2), seasonal = list(order = c(0,1,1), period = 12))
fcast_112 <- forecast(sarima_112, h = 24)

# Forecast data
fcast_df_112 <- data.frame(
  Date = as.Date(as.yearmon(time(fcast_112$mean))),
  Forecast = as.numeric(fcast_112$mean),
  Lower = as.numeric(fcast_112$lower[, 2]),
  Upper = as.numeric(fcast_112$upper[, 2])
)

# Plot
ggplot() +
  geom_line(data = full_df, aes(x = Date, y = Value), color = "black") +
  geom_ribbon(data = fcast_df_112, aes(x = Date, ymin = Lower, ymax = Upper), fill = "blue", alpha = 0.5) +
  geom_line(data = fcast_df_112, aes(x = Date, y = Forecast), color = "red") +

  ggtitle("SARIMA(1,1,2)(0,1,1)[12] - Trained on 1949-1958") +
  ylab("Residuals") + xlab("Year") +
  theme_minimal()
```



```
library(forecast)
library(zoo)

log_ap_ts <- log(AirPassengers)

time_vals_log <- time(log_ap_ts)
harmonics <- harmonic(ts(log_ap_ts, frequency = 12), m = 2)
colnames(harmonics) <- c("cos1", "cos2", "sin1", "sin2")

df <- data.frame(
  log_ap_ts = log_ap_ts,
  time_vals_log = time_vals_log,
  harmonics
)

harm_model_log_k2 <- lm(log_ap_ts ~ time_vals_log + cos1 + cos2 + sin1 + sin2, data = df)

resid_log_k2 <- residuals(harm_model_log_k2)
resid_log_k2_ts <- ts(resid_log_k2, start = c(1949, 1), frequency = 12)

final_model <- Arima(resid_log_k2_ts, order = c(2,1,1),
  seasonal = list(order = c(0,1,1), period = 12))

resid_forecast <- forecast(final_model, h = 12)
```

```

future_time_vals_log <- time_vals_log[length(time_vals_log)] + (1:12)/12
future_ts <- ts(rep(0, 12), start = c(1961, 1), frequency = 12)
future_harm <- harmonic(future_ts, m = 2)
colnames(future_harm) <- c("cos1", "cos2", "sin1", "sin2")

newdata <- data.frame(
  time_vals_log = future_time_vals_log,
  future_harm
)

future_trend <- predict(harm_model_log_k2, newdata = newdata)
log_forecast_full_model <- future_trend + resid_forecast$mean

log_forecast_ts <- ts(log_forecast_full_model, start = c(1961, 1), frequency = 12)

plot(log_ap_ts, xlim = c(1949, 1962), ylim = range(log_ap_ts, log_forecast_ts),
     main = "Final Forecast (Log Scale) - Harmonic + SARIMA",
     ylab = "log(Passengers)", xlab = "Year")

lines(log_forecast_ts, col = "red", lwd = 2)
points(log_forecast_ts, col = "red", pch = 19)

legend("topleft", legend = c("Observed (log)", "Forecast (log)"),
     col = c("black", "red"), lty = 1, lwd = 2)

```

Final Forecast (Log Scale) – Harmonic + SARIMA

